A Two-Sided Bayes-Error Bracket from Partition-Conditional Entropy

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Abstract

Many machine-learning predictors are constant on the cells of a fixed partition of their input space: decision trees on their leaves, vector quantisers and codebook classifiers on Voronoi cells, message-passing graph neural networks on Weisfeiler–Leman colour classes. For all such predictors the partition is the irreducible expressivity bottleneck. We give one elementary two-sided closed-form bracket, in the form ML practitioners need, between the *partition-restricted minimum (empirical) risk* ε^*_Π of the best constant-on-cells predictor and the partition-conditional entropy $H(f \mid \Pi)$ of a binary label f:

$$H_{\mathrm{bin}}^{-1}\big(H(f \mid \Pi)\big) \;\le\; \varepsilon^*_\Pi \;\le\; \tfrac{1}{2}\, H(f \mid \Pi).$$

The bracket is tight on both boundaries by explicit witnesses, unimprovable in closed form (no constant smaller than 1/2, no pointwise-larger lower function), and pins ε^*_Π to a window of width at most $w^* \approx 0.1610$ uniformly in Π and f. Three worked applications — decision trees, vector quantisation, and message-passing networks via Weisfeiler–Leman — show how the bracket lifts to architecture-independent, training-free error bounds. A short companion Julia script verifies the bracket on 1,000 random partitions in exact rational arithmetic (for ε^*_Π) together with certified interval arithmetic (for the Shannon-entropy terms); zero violations, and the empirically observed maximum upper-side slack reproduces w^* to four decimals. On real benchmarks the bracket is two orders of magnitude cheaper than a single training epoch, on tabular data the empirical Bayes floor matches CART and logistic regression to four decimals at every cell budget tested, and on featureless structural-WL benchmarks (Cayley and Paley graphs) the bracket correctly pins at the marginal-entropy ceiling on the tasks 1-WL is provably blind to — evidence that the diagnostic has both the right success regime and the right *failure* regime, the latter being the empirical face of an ε-robust refinement of the classical MPNN–WL constancy lemma that we prove and exercise here.

1 Introduction

From inputs to vertices. Pick any supervised learning problem with a finite or finitely representable input set: a training corpus of n examples, the nodes of a graph, the pixels of an image grid, the leaves of a decision tree, the codewords of a quantiser, the time-steps of a discretised process. Call the elements of this set *vertices* and write the set V. A binary classification task on V is then nothing but a function $f : V \to \{0, 1\}$ — the *label vector* we wish to predict. This is the most elementary supervised-learning datum: "which vertices are positive?".

What a model actually sees. A learning algorithm rarely processes V directly. Instead, the architecture — a tree depth, a codebook size, a network depth, a feature map — imposes an *equivalence relation* on V: two vertices look identical to the model if and only if they route to the same leaf, the same codeword, the same Weisfeiler–Leman colour, or the same feature-map output. The quotient of V by this equivalence relation is a *partition* $\Pi = \{C_1, \dots, C_m\}$ of V into nonempty *cells* $C_i \subset V$. The architecture-imposed map $C : V \to \Pi$, sending each vertex to its cell, is the *lens* through which the model perceives the input. Anything the model predicts about a vertex v must factor as

$$\widehat{f}(v) \;=\; g\big(C(v)\big), \qquad g : \Pi \to \{0,1\}.$$

This factorisation is exact, not approximate: it is forced by the architecture before training begins. In particular, two vertices in the same cell must receive the same predicted label, *regardless of how much training data, parameters, or epochs are spent.* The partition Π is therefore the **irreducible expressivity bottleneck** of the model family.

Three concrete instances. The vertex–partition view unifies models that look superficially unrelated:

- **Decision trees and random forests.** V is the training set; cells C_i are the leaves; the equivalence "$v \sim w$ iff v and w end at the same leaf" is the partition Π_T. Every leaf-constant prediction rule of CART, C4.5 or any greedy tree learner factors through Π_T (Breiman et al., 1984; Devroye et al., 1996).
- **Vector quantisers and clustering classifiers.** V is the input space (or a finite training sample); cells are pre-images of the k codewords (Voronoi cells in the metric case). Nearest-codeword classification, k-means classifiers and product-quantisation retrieval all factor through their codebook partition Π_Q (Gersho & Gray, 1991; Jégou et al., 2011).
- **Message-passing graph neural networks (MPNNs).** V is the vertex set of a graph G; the depth-L Weisfeiler–Leman colour refinement $\mathrm{WL}_L(G)$ is a partition of V. Every depth-L MPNN with shared, permutation-invariant updates and no node identifiers is constant on the cells of $\mathrm{WL}_L(G)$ (Xu et al., 2019; Morris et al., 2019): WL is the lens through which a bounded-depth MPNN sees its graph.

In each case, the architectural choice (tree depth, codebook size, MPNN depth) selects a partition; the partition determines what the model can possibly express; the label f is the only thing the model is asked to fit. The triple (V, Π, f) is the data of the problem.

The two questions. Once one accepts this lens, the practical questions of supervised learning collapse to two:

(Q1) *Lower bound — no predictor can do better.* Given a partition Π of V and a label $f : V \to \{0,1\}$, what is the smallest possible error achievable by *any* function $g : \Pi \to \{0,1\}$? We call this the *partition-restricted minimum (empirical) risk* ε^*_Π, deliberately avoiding the term "Bayes error" which in statistical learning theory denotes the population infimum over *all* measurable classifiers. Here ε^*_Π is the minimum risk on V over the architecture-restricted hypothesis class $\{g \circ C : g : \Pi \to \{0,1\}\}$, i.e. an empirical quantity under a structural constraint.

(Q2) *Upper bound — some predictor achieves it.* Is there a quantity, easily computed from (Π, f), that *certifies* a small ε^*_Π from above? Such a certificate would let a practitioner read off a guaranteed error level without ever training a model.

Why entropy? The natural functional of a label $f : V \to \{0,1\}$ relative to a partition Π is the *partition-conditional Shannon entropy*

$$H(f \mid \Pi) \;:=\; \sum_{C \in \Pi} q_C \cdot H_{\mathrm{bin}}(P_C),$$

where q_C is the mass of cell C and P_C is the fraction of positives inside C. The quantity $H(f \mid \Pi)$ measures the average *impurity* of the label within cells: it is zero exactly when every cell is monochromatic ($P_C \in \{0,1\}$ for every C), and reaches 1 exactly when every cell is maximally mixed ($P_C = 1/2$ for every C). Three reasons make $H(f \mid \Pi)$ inevitable as the right diagnostic:

- *Decision trees already compute it.* CART's information-gain splitting criterion is the change in $H(f \mid \Pi)$ when one refines Π at a candidate split. The bracket below converts that internal criterion into an external error guarantee.
- *It is sub-σ-algebra entropy.* In the measure-theoretic view, Π generates a sub-σ-algebra of the power set of V, and $H(f \mid \Pi)$ is the standard conditional entropy of the random label f given this sub-σ-algebra (Cover & Thomas, 2006, §2.1). Every partition-respecting predictor is measurable with respect to this sub-σ-algebra; conditional entropy is the canonical information-theoretic obstruction to recovering f from such predictors.
- *It is the only functional with a two-sided bracket.* Classical inequalities of Fano (1961) and Hellman & Raviv (1970) relate ε^*_Π to $H(f \mid \Pi)$ from both sides; no other elementary scalar functional of (Π, f) enjoys both a matching lower bound *and* a matching upper bound in closed form.

This paper. For binary labels we answer (Q1) and (Q2) simultaneously by a single classical inequality recast for partitions: the partition-conditional Shannon entropy $H(f \mid \Pi)$ two-sidedly brackets ε^*_Π:

$$H_{\mathrm{bin}}^{-1}\big(H(f \mid \Pi)\big) \;\le\; \varepsilon^*_\Pi \;\le\; \tfrac{1}{2}\, H(f \mid \Pi).$$

The lower side specialises Fano's inequality (Fano, 1961); the upper side specialises the Hellman–Raviv bound (Hellman & Raviv, 1970); the two-sided packaging is in the spirit of Feder & Merhav (1994) and the modern treatment of Ho & Verdú (2010). The contribution of this paper is not the inequalities themselves but their packaging as a *single architecture-level diagnostic* on (V, Π, f): once computed, the bracket gives the practitioner *simultaneously* an unconditional lower bound on the error of every model in the architectural class, and a constructive upper bound achieved by the elementary plug-in predictor on cells.

What the bracket enables. A practitioner armed with $H(f \mid \Pi)$ for a candidate architecture can:

(E1) *Reject architectures cheaply (the underfitting floor).* If $H_{\mathrm{bin}}^{-1}(H(f \mid \Pi))$ exceeds the target error on V, no training run of any model in the family can fit the data below that target; reject the family before spending compute. This is a *one-sided* diagnostic: a high empirical-risk floor is necessarily inherited by any population risk (training cannot beat its own ceiling).

(E2) *Certify architectural capacity cheaply.* If $\frac{1}{2}H(f \mid \Pi)$ is smaller than the target error, the plug-in predictor on cells already meets the target *on* V, with no optimiser, no validation set, and no hyperparameters. This certifies that the architecture has enough expressive capacity on V; it does *not* by itself guarantee out-of-sample generalisation, which requires the population extension of Theorem 24.

(E3) *Compare architectures on equal footing.* Two different partitions Π_1, Π_2 (say, tree depths or MPNN depths) yield two brackets; the one with smaller $H(f \mid \Pi)$ dominates uniformly in the optimiser.

(E4) *Track refinement gains.* If Π' refines Π (deeper tree, larger codebook, more MPNN layers), $H(f \mid \Pi') \le H(f \mid \Pi)$ and both bracket endpoints move monotonically (Theorem 11). This quantifies the marginal value of model-class growth.

Cost and scope. "Training-free" here means *optimisation-free*: computing the bracket requires the labels f (so the diagnostic is supervised, not unsupervised), but it requires no gradient steps, parameter initialisation, learning-rate tuning, or validation set. The cost of evaluating $H(f \mid \Pi)$ is a single $O(|V|)$ pass to accumulate cell frequencies P_C and masses q_C; this is strictly dominated by the cost of one training epoch. On four binary tabular benchmarks ranging from $n = 178$ to $n = 45{,}222$ we measure (Table 12) that the bracket is between $20\times$ and $133\times$ cheaper than fitting a single CART tree of the same cell budget, and cheaper than one LBFGS epoch of logistic regression on three of the four datasets.

The bracket is tight on both boundaries by explicit witnesses, and we show in Theorem 13 that no closed-form improvement is possible. The width of the bracket is bounded above by an explicit constant $w^* \approx 0.1610$ uniformly in Π and f — to our knowledge the first *uniform conservativeness gap* reported in the entropy-bounds-Bayes-error literature.

Contributions.

(C1) *Bracket.* A self-contained two-sided closed-form bracket (Theorem 3) on the partition-restricted minimum risk of every partition-constant predictor of a binary label, with an elementary finite-sum proof.

(C2) *Universal slack constant.* The explicit constant $w^* = \frac{1}{2}H_{\text{bin}}(1/5) - 1/5 \approx 0.1610$ (Theorem 5), bounding the bracket's width uniformly over *all* finite partitions and *all* binary labels. To our knowledge this is the first such uniform conservativeness gap for the entropy/error relation, and it is the central theoretical novelty of the paper.

(C3) *Marginal-aware refinement.* A sharper bracket (Theorem 6) using the global label marginal π: the marginal-aware slack $w^*(\pi_*)$ shrinks strictly below w^* for unbalanced labels, e.g. $w^*(0.1) \approx 0.069$.

(C4) *Empirical-to-population extension.* A standard concentration upgrade (Theorem 24) converting the bracket on V into a PAC-style population diagnostic with an $O\big(\sqrt{\log m/n}\big)$ additive correction under bounded cell mass and bounded cell mean.

(C5) *Sharpness.* A negative result (Theorem 13) ruling out any uniform improvement of either side by simpler closed-form expressions.

(C6) *Three worked applications.* Decision trees, vector quantisation, and Weisfeiler–Leman-refined message passing (§ 9), each instantiated with numerical examples, yielding training-free error brackets for that family.

(C7) *Mechanised check.* A short Julia script (`verify.jl`, shipped with this preprint) verifies the bracket on 10^3 random partitions in exact rational arithmetic (for ε^*_Π) together with certified interval arithmetic (for the Shannon-entropy terms); zero violations.

(C8) *Empirical evidence on real benchmarks.* Eight self-contained experiments (§ 9.5) on UCI Adult, breast-cancer, wine, digits-binary, Twitch-EN, Cora, CiteSeer, PubMed, ogbn-arxiv, and a featureless structural-WL battery (Cayley + Paley). Highlights: on UCI Adult, CART training error matches $\varepsilon^*_{\Pi_d}$ to four decimal places at every depth $d = 1, \dots, 15$ (Table 2) and logistic-regression-on-cells matches $\varepsilon^*_{\Pi_k}$ for every $k \le 256$ (Table 3); per-arch bracket evaluation is 20–133× cheaper than one CART fit (Table 12); as a heterogeneous-menu NAS pre-filter the bracket attains Kendall $\tau = 0.48$ ($p = 5\times10^{-5}$) with held-out test error on Adult while parameter count *anti*-ranks the same menu ($\tau = -0.38$, Table 11); on feature-rich citation / social graphs the depth-3 1-WL partition becomes near-discrete (95.6% singleton cells on ogbn-arxiv), exposing the bracket's partition-cardinality collapse regime which we report transparently rather than claim as a victory (Table 5); on featureless structural-WL benchmarks the bracket correctly pins at the marginal-entropy ceiling (Table 6); empirical concentration to the population bracket matches the $\Theta(1/\sqrt{n})$ rate of Theorem 24 with 1.000 coverage at the nominal 0.95 level (Table 13).

Scope. All statements are for finite V, finite partition Π, and binary f. Multi-class extensions ($|\mathcal{Y}| > 2$) proceed via the concave-envelope construction of Feder & Merhav (1994) and are not pursued here. Continuous-alphabet generalisations require additional regularity (Ho & Verdú, 2010) and are also not pursued.

Related work. See § 10 for placement against the entropy-bounds-error literature (Fano, 1961; Hellman & Raviv, 1970; Tebbe & Dwyer III, 1968; Kovalevskij, 1968; Feder & Merhav, 1994; Han & Verdú, 1994; Ho & Verdú, 2010; Sason & Verdú, 2018), the message-passing expressivity literature (Xu et al., 2019; Morris et al., 2019; Böker et al., 2023), and decision-tree / VQ foundations (Breiman et al., 1984; Gersho & Gray, 1991; Devroye et al., 1996).

2 Setup

We now make the vertex–partition–label triple (V, Π, f) of § 1 formal, and fix the two quantities that the rest of the paper relates: the partition-restricted minimum risk ε^*_Π (the quantity to be bounded) and the partition-conditional entropy $H(f \mid \Pi)$ (the bound).

Let V be a finite set with $|V| = n \geq 1$, and let Π be a partition of V into nonempty cells $\Pi = \{C_1, \dots, C_m\}$. Let $f : V \to \{0,1\}$ be a binary label, and write $C(v)$ for the unique cell of Π containing $v \in V$. Endow V with the uniform probability measure $q_v = 1/n$ (Theorem 2 discusses non-uniform weights).

Cell statistics. For each cell $C \in \Pi$ write

$$q_C := |C|/n, \qquad P_C := |C|^{-1} \sum_{v \in C} f(v), \qquad e_C := \min(P_C, 1 - P_C) \;\in\; [0, 1/2].$$

Thus q_C is the mass of cell C, P_C is the cell-conditional mean of f, and e_C is the within-cell majority-prediction error.

Partition-restricted minimum risk. The minimum error of any constant-on-cells predictor is

$$\varepsilon^*_\Pi \;:=\; \min_{g:\Pi \to \{0,1\}} \mathbb{P}\big[g(C(v)) \neq f(v)\big] \;=\; \sum_{C \in \Pi} q_C \cdot e_C, \tag{1}$$

attained by the plug-in rule $\hat{h}_\Pi(v) := \mathbf{1}\{P_{C(v)} \geq 1/2\}$. Equivalently, ε^*_Π is the Bayes risk of f when the predictor is restricted to depend on v only through $C(v)$ (Devroye et al., 1996, §2.1).

Partition-conditional entropy. In bits,

$$H(f \mid \Pi) \;:=\; \sum_{C \in \Pi} q_C \cdot H_{\mathrm{bin}}(P_C) \;=\; \sum_{C \in \Pi} q_C \cdot H_{\mathrm{bin}}(e_C),$$

where $H_{\mathrm{bin}}(p) := -p \log_2 p - (1-p)\log_2(1-p)$ with $0 \log_2 0 := 0$, and the second equality is the symmetry $H_{\mathrm{bin}}(p) = H_{\mathrm{bin}}(1-p)$.

Inverse binary entropy. The function $H_{\mathrm{bin}} : [0,1] \to [0,1]$ is strictly concave, symmetric about $1/2$, with maximum $H_{\mathrm{bin}}(1/2) = 1$, strictly increasing on $[0, 1/2]$. Its inverse on the increasing branch is the unique

$$H_{\mathrm{bin}}^{-1} : [0,1] \to [0, 1/2]$$

satisfying $H_{\mathrm{bin}}(H_{\mathrm{bin}}^{-1}(h)) = h$. As the inverse of a concave increasing function, H_{bin}^{-1} is *convex*, continuous, and strictly increasing, with $H_{\mathrm{bin}}^{-1}(0) = 0$ and $H_{\mathrm{bin}}^{-1}(1) = 1/2$.

Remark 1 (range of $H(f \mid \Pi)$). Since $H_{\mathrm{bin}}(P_C) \in [0,1]$ for every C and $\sum_C q_C = 1$, the partition-conditional entropy satisfies $H(f \mid \Pi) \in [0,1]$ in bits. In particular, $H(f \mid \Pi)$ lies in the domain of H_{bin}^{-1}, justifying the left-hand side of (2) below.

Remark 2 (non-uniform weighting). All statements and proofs below extend verbatim to non-uniform weights $\{q_v\}_{v \in V}$ with $\sum_v q_v = 1$, by redefining $q_C := \sum_{v \in C} q_v$ and $P_C := q_C^{-1} \sum_{v \in C} q_v f(v)$. The only properties of $\{q_C\}$ used are nonnegativity and $\sum_C q_C = 1$.

3 Main result

Theorem 3 (Two-sided bracket)**.** *For every finite partition Π of a finite set V and every binary $f : V \to \{0,1\}$,*

$$H_{\mathrm{bin}}^{-1}\big(H(f \mid \Pi)\big) \;\leq\; \varepsilon^*_\Pi \;\leq\; \tfrac{1}{2} H(f \mid \Pi). \tag{2}$$

The two inequalities are respectively the lower and upper bounds of Ho & Verdú (2010) specialised to the partition channel, in turn refining classical results of Fano (1961) and Hellman & Raviv (1970) packaged in the achievable-region form of Feder & Merhav (1994). The contribution of this paper is not the theorem itself but its packaging and consequences (Theorems 5 and 13 and § 9).

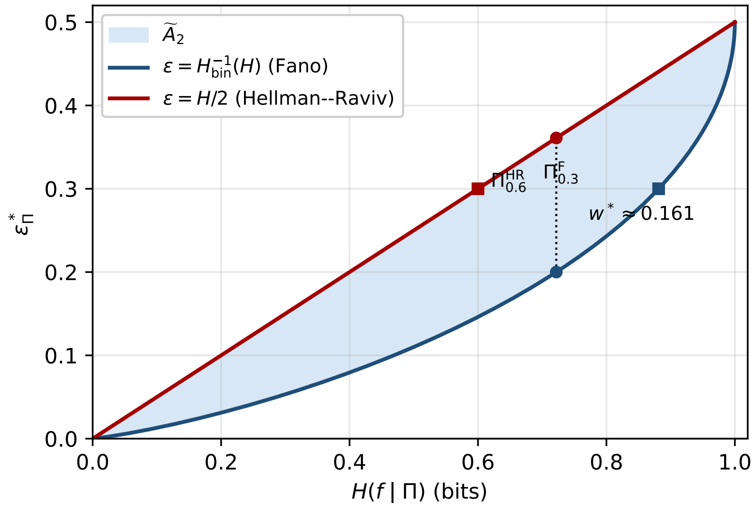

Figure 1: The achievable region $\widetilde{A}_2$ on the (H, ε) plane. Upper boundary $\varepsilon = H/2$ (Hellman–Raviv, tight via Π^{HR}_α). Lower boundary $\varepsilon = H_{\mathrm{bin}}^{-1}(H)$ (Fano, tight via $\Pi^{\mathrm{F}}_\varepsilon$). Maximum width $w^* \approx 0.1610$ attained at $\varepsilon = 1/5$. Every $(\varepsilon^*_\Pi, H(f \mid \Pi))$ from a finite partition and binary label lies in the shaded region; both boundaries and the interior are achievable.

Achievable region. Define the candidate region

$$\widetilde{A}_2 \;:=\; \left\{ (\varepsilon, H) \in [0, 1/2] \times [0, 1] \;:\; H_{\mathrm{bin}}^{-1}(H) \le \varepsilon \le H/2 \right\},$$

shown in Figure 1.

Proposition 4 (Achievable region)**.** *$\widetilde{A}_2$ is exactly the closure of the set of pairs $(\varepsilon^*_\Pi, H(f \mid \Pi))$ achievable by some finite partition Π of a finite set V and some binary label $f : V \to \{0, 1\}$. Concretely:*

(i) Every achievable pair lies in $\widetilde{A}_2$ (Theorem 3).

(ii) The lower boundary $\{(\varepsilon, H_{\mathrm{bin}}(\varepsilon)) : \varepsilon \in [0, 1/2]\}$ is realised by the trivial-partition Fano family $\Pi^{\mathrm{F}}_\varepsilon$ of § 8.

(iii) The upper boundary $\{(H/2, H) : H \in [0, 1]\}$ is realised by the two-cell Hellman–Raviv family Π^{HR}_α of § 8 with $\alpha = H$.

(iv) Every interior point of $\widetilde{A}_2$ is realised by a disjoint union of one boundary witness with appropriate mass split (Theorem 15).

The proof is collected after the boundary witnesses are defined in § 8; see Theorem 15 for the explicit interior construction.

Corollary 5 (Uniform slack)**.** *Let $w(H) := \frac{1}{2}H - H_{\mathrm{bin}}^{-1}(H)$ be the width of the bracket (2) at conditional entropy H. Then w vanishes at $H \in \{0, 1\}$, is concave on $[0, 1]$, and attains its maximum*

$$w^* \;:=\; \max_{H \in [0,1]} w(H) \;=\; \tfrac{1}{2}\, H_{\mathrm{bin}}(1/5) - 1/5 \;\approx\; 0.1610$$

at $H^* = H_{\mathrm{bin}}(1/5) \approx 0.7219$, equivalently at $\varepsilon = 1/5$. Consequently, for every Π and f,

$$\tfrac{1}{2}\, H(f \mid \Pi) - \varepsilon^*_\Pi \;\le\; w^* \quad \textit{and} \quad \varepsilon^*_\Pi - H_{\mathrm{bin}}^{-1}(H(f \mid \Pi)) \;\le\; w^*.$$

Proof. $w'(H) = \frac{1}{2} - 1/H'_{\mathrm{bin}}(H_{\mathrm{bin}}^{-1}(H))$ by the chain rule. Setting $w'(H) = 0$ gives $H'_{\mathrm{bin}}(\varepsilon) = 2$, equivalently $\log_2((1-\varepsilon)/\varepsilon) = 2$, i.e. $\varepsilon = 1/5$. Substituting back gives $H^* = H_{\mathrm{bin}}(1/5)$ and the value w^*. Concavity of w follows from concavity of $H/2$ (linear) and convexity of H_{bin}^{-1} (so $-H_{\mathrm{bin}}^{-1}$ is concave). Vanishing at the endpoints is immediate. $\square$

4 Marginal-aware refinement

The slack of Theorem 5 is uniform in Π and f, but it is loose when the global label marginal is far from $1/2$: no constant-on-cells predictor can possibly do worse than the trivial *global majority* predictor, and incorporating this trivial upper bound tightens the upper side of (2) for unbalanced labels.

Proposition 6 (Marginal-aware bracket)**.** *Let* $\pi := \sum_{v \in V} q_v f(v) \in [0,1]$ *be the global mean of* f*, where* $q_v := 1/n$*, and write* $\pi_* := \min(\pi, 1-\pi) \in [0, 1/2]$*. Then*

$$H_{\mathrm{bin}}^{-1}\big(H(f \mid \Pi)\big) \;\le\; \varepsilon_{\Pi}^{*} \;\le\; \min\Big(\tfrac{1}{2}H(f \mid \Pi),\; \pi_*\Big). \tag{3}$$

The associated marginal-aware slack

$$w^*(\pi_*) \;:=\; \max_{H \in [0,1]} \big[\min(\tfrac{1}{2}H, \pi_*) - H_{\mathrm{bin}}^{-1}(H)\big]$$

admits the closed form

$$w^*(\pi_*) \;=\; \begin{cases} w^* \;\approx\; 0.1610, & \pi_* \ge \tfrac{1}{2}H_{\mathrm{bin}}(1/5) \;\approx\; 0.3610, \\ \pi_* - H_{\mathrm{bin}}^{-1}(2\pi_*), & \pi_* < \tfrac{1}{2}H_{\mathrm{bin}}(1/5), \end{cases} \tag{4}$$

and satisfies $w^*(\pi_*) \le w^*$ *with equality iff* $\pi_* \ge \tfrac{1}{2}H_{\mathrm{bin}}(1/5)$*.*

Proof. The constant predictor $g \equiv \mathbf{1}\{\pi \ge 1/2\}$ on V has error rate π_* and is trivially constant on every cell, hence $\varepsilon_{\Pi}^{*} \le \pi_*$. Combining with Theorem 3 gives (3).

For the slack, let $\phi(H) := \min(\tfrac{1}{2}H, \pi_*) - H_{\mathrm{bin}}^{-1}(H)$. On $[0, 2\pi_*]$, $\phi(H) = w(H) = \tfrac{1}{2}H - H_{\mathrm{bin}}^{-1}(H)$, the function of Theorem 5, which is concave on $[0,1]$ with unique critical point $H^* = H_{\mathrm{bin}}(1/5)$ and maximal value w^*. On $[2\pi_*, 1]$, $\phi(H) = \pi_* - H_{\mathrm{bin}}^{-1}(H)$ is strictly decreasing in H, so the maximum of ϕ over $[2\pi_*, 1]$ is attained at $H = 2\pi_*$ with value $\pi_* - H_{\mathrm{bin}}^{-1}(2\pi_*)$. Two cases:

- If $2\pi_* \ge H^*$, i.e. $\pi_* \ge \tfrac{1}{2}H_{\mathrm{bin}}(1/5)$, the global maximum of w on $[0,1]$ lies inside $[0, 2\pi_*]$, so $w^*(\pi_*) = w^*$.
- If $2\pi_* < H^*$, then w is strictly increasing on $[0, 2\pi_*]$ (as H^* is its unique critical point and $w(0) = 0$), so the maximum over $[0, 2\pi_*]$ is at $H = 2\pi_*$ with value $\pi_* - H_{\mathrm{bin}}^{-1}(2\pi_*)$, which also matches the right-branch value, giving $w^*(\pi_*) = \pi_* - H_{\mathrm{bin}}^{-1}(2\pi_*)$.

The inequality $w^*(\pi_*) \le w^*$ is then immediate. ☐

Theorem 6 reproduces the universal slack $w^* \approx 0.1610$ at $\pi_* = 1/2$ and shrinks it substantially for unbalanced tasks; Table 1 reports the closed form on a grid. A symbolic verification of (4) and the table is included in the extended T3 script (§ 9.4).

π_*	$w^*(\pi_*)$	π_*	$w^*(\pi_*)$
0.05	0.0370	0.25	0.1400
0.10	0.0689	0.30	0.1539
0.15	0.0968	0.35	0.1607
0.20	0.1206	0.50	0.1610

Table 1: Marginal-aware slack $w^*(\pi_*)$ from Theorem 6, computed from (4). The plateau at $\pi_* \ge 0.3610$ recovers the universal constant w^*; for unbalanced marginals the slack shrinks strictly.

Remark 7 (Towards prior-aware lower bounds). Theorem 6 tightens only the *upper* side via the trivial-majority observation. A sharper *lower* side under known marginals is available via the data-processing refinements of Han & Verdú (1994) and Ho & Verdú (2010); we defer the fully prior-aware bracket to a companion paper.

5 Proof of Theorem 3

The proof reduces to one scalar inequality.

Lemma 8 (Scalar Hellman–Raviv). *For every $p \in [0,1]$,*

$$\min(p, 1-p) \;\le\; \tfrac{1}{2}\, H_{\mathrm{bin}}(p), \tag{5}$$

with equality iff $p \in \{0, 1/2, 1\}$.

Proof. By the symmetry $p \mapsto 1-p$ both sides are invariant, so we may assume $p \in [0,1/2]$ and (5) becomes $g(p) := \frac{1}{2}H_{\mathrm{bin}}(p) - p \ge 0$. The function g is concave on $[0,1/2]$ (sum of the concave $\frac{1}{2}H_{\mathrm{bin}}$ and the linear $-p$) with $g(0) = g(1/2) = 0$; hence $g \ge 0$ on $[0,1/2]$. Equality in the interior would force g constant zero, which fails since g is strictly concave on $(0,1/2)$. □

Proof of Theorem 3. Upper bound. Apply (5) with $p = P_C$ to each cell and multiply by $q_C \ge 0$: $q_C \cdot e_C \le \frac{1}{2}\, q_C \cdot H_{\mathrm{bin}}(P_C)$. Summing over $C \in \Pi$ gives $\varepsilon^*_\Pi \le \frac{1}{2}\, H(f \mid \Pi)$.

Lower bound. The symmetry $H_{\mathrm{bin}}(P_C) = H_{\mathrm{bin}}(e_C)$ yields $H(f \mid \Pi) = \sum_C q_C H_{\mathrm{bin}}(e_C)$. Since H_{bin} is concave on $[0,1]$ and $\{e_C\} \subset [0,1/2]$, Jensen's inequality with weights $\{q_C\}_C$ gives

$$H(f \mid \Pi) \;=\; \sum_C q_C\, H_{\mathrm{bin}}(e_C) \;\le\; H_{\mathrm{bin}}\Big(\sum_C q_C\, e_C\Big) \;=\; H_{\mathrm{bin}}(\varepsilon^*_\Pi).$$

Since $\varepsilon^*_\Pi \in [0,1/2]$ and H_{bin}^{-1} is the inverse of H_{bin} on the increasing branch $[0,1/2]$, applying H_{bin}^{-1} to both sides yields $H_{\mathrm{bin}}^{-1}(H(f \mid \Pi)) \le \varepsilon^*_\Pi$. □

Remark 9 (Lagrangian dual programs). Both halves of (2) can be read as opposite extremal programs on the cell statistics $\{(q_C, e_C)\}_C$: the lower bound equals $\sup\{\sum_C q_C H_{\mathrm{bin}}(e_C) : q_C \ge 0, \sum_C q_C = 1, \sum_C q_C e_C = \varepsilon\} = H_{\mathrm{bin}}(\varepsilon)$ (maximised at the constant interior point $e_C \equiv \varepsilon$, Jensen), the upper bound equals $\sup\{\sum_C q_C e_C : q_C \ge 0, \sum_C q_C = 1, \sum_C q_C H_{\mathrm{bin}}(e_C) = H\} = H/2$ (maximised on the boundary set $e_C \in \{0, 1/2\}$, Theorem 8). The Feder–Merhav primal concave-envelope construction (Feder & Merhav, 1994) traces the upper boundary of $\widetilde{A}_2$ by exactly this boundary mixture; the dual programs give the optimal value in one line each.

Remark 10 (Gini variance form). A useful consequence of (5) is the variance inequality $\varepsilon^*_\Pi \le 2\,\mathbb{E}[\mathrm{Var}(f \mid \Pi)] = 2\sum_C q_C P_C(1-P_C)$, obtained from the elementary $\min(p, 1-p) \le 2p(1-p)$ for $p \in [0,1]$. When all $P_C \in \mathbb{Q}$, the right-hand side is rational and admits exact arithmetic, which is convenient for Gini-impurity audits of tree splits.

6 Refinement monotonicity

A partition Π' *refines* Π, written $\Pi' \preceq \Pi$, if every cell of Π' is contained in some cell of Π. Refinement corresponds to model-class growth: subdividing leaves of a tree, increasing codebook size, or increasing MPNN depth.

Proposition 11 (Refinement monotonicity). *If $\Pi' \preceq \Pi$, then for every binary f,*

$$H(f \mid \Pi') \le H(f \mid \Pi), \qquad \varepsilon^*_{\Pi'} \le \varepsilon^*_\Pi.$$

Proof. Fix $C \in \Pi$ and write $C = \bigsqcup_j C'_j$ in Π' with masses $q_{C'_j}$ summing to q_C. The cell mean P_C is the convex combination $\sum_j (q_{C'_j}/q_C) P_{C'_j}$, so by concavity of H_{bin} on $[0,1]$, $\sum_j q_{C'_j} H_{\mathrm{bin}}(P_{C'_j}) \le q_C H_{\mathrm{bin}}(P_C)$. Summing over C gives the entropy claim. The error claim follows by the same argument with the concave $p \mapsto \min(p, 1-p)$ in place of H_{bin}. □

This is the deterministic-coarsening case of the data-processing inequality (Cover & Thomas, 2006, Thm. 2.8.1). We use it in § 9 to track how the bracket tightens under model-class growth.

Proposition 12 (Refinement to the discrete partition)**.** *Let $\Pi_0 \preceq \Pi_1 \preceq \cdots \preceq \Pi_L$ be a chain of partitions of a finite V with Π_L the* discrete *partition $\{\{v\} : v \in V\}$. For every binary label $f : V \to \{0,1\}$,*

$$H(f \mid \Pi_L) \;=\; 0, \qquad \varepsilon^*_{\Pi_L} \;=\; 0,$$

so both endpoints of the bracket (2) collapse to 0 at level L. Consequently the bracket width $\frac{1}{2}H(f \mid \Pi_\ell) - H_{\text{bin}}^{-1}(H(f \mid \Pi_\ell))$ is monotonically non-increasing in ℓ along the chain and tends to 0 as Π_ℓ approaches discreteness.

Proof. On a singleton cell $\{v\}$, $P_{\{v\}} = f(v) \in \{0,1\}$, so $H_{\text{bin}}(P_{\{v\}}) = 0$ and $\min(P_{\{v\}}, 1 - P_{\{v\}}) = 0$; summing over v gives the two equalities. Monotonicity of the bracket endpoints along the chain is Theorem 11; combined with $H_{\text{bin}}^{-1}(0) = 0$ and $\frac{1}{2} \cdot 0 = 0$, the bracket width vanishes. ☐

Theorem 12 is a consistency check, not a positive result: it certifies that the bracket reports *exactly zero* on a partition refined to singletons, which is the correct value of the Bayes error (any deterministic labelling has zero training-set risk on its own preimage). This makes the empirically observed pinch at the discrete-partition extreme (Table 5, ogbn-arxiv at depth 3 with 95.6% singleton cells) a *verification* of the bracket's asymptotic behaviour rather than a failure mode. The informative regime is the intermediate one, where $|\Pi_\ell|/|V|$ is bounded away from both 0 (trivial partition) and 1 (discrete partition); the bracket width is a quantitative diagnostic for how far Π_ℓ has progressed along the chain.

7 Sharpness

Proposition 13 (Unimprovability)**.**

(i) For every constant $c < 1/2$ there exists a finite partition Π of a finite V and a binary f with $\varepsilon^*_\Pi > c \cdot H(f \mid \Pi)$. In particular, no constant smaller than $1/2$ can replace the coefficient on the right-hand side of (2).

(ii) For every function $\psi : [0,1] \to [0,1/2]$ such that the set $\{H : \psi(H) > H_{\text{bin}}^{-1}(H)\}$ contains an open subinterval $U \subset (0,1)$, there exists a finite partition Π of a finite V and a binary f with $\varepsilon^*_\Pi < \psi(H(f \mid \Pi))$.

Proof. (i) The upper-boundary family Π^{HR}_α of § 8 achieves $\varepsilon^*_\Pi = \frac{1}{2}H(f \mid \Pi)$ for every $\alpha \in (0,1]$; any $c < 1/2$ is witnessed by such a row.

(ii) Since H_{bin} is a continuous bijection $[0,1/2] \to [0,1]$, the image $H_{\text{bin}}^{-1}(U)$ contains an open subinterval of $(0,1/2)$. Pick a rational $\varepsilon \in H_{\text{bin}}^{-1}(U) \cap (0,1/2)$ and a positive integer n with $\varepsilon n \in \mathbb{Z}$. The lower-boundary family $\Pi^{\text{F}}_\varepsilon$ of § 8 on this n satisfies $\varepsilon^*_\Pi = \varepsilon = H_{\text{bin}}^{-1}(H(f \mid \Pi))$ and $H(f \mid \Pi) = H_{\text{bin}}(\varepsilon) \in U$, so $\varepsilon^*_\Pi < \psi(H(f \mid \Pi))$. ☐

Together, Theorem 3 and Theorem 13 identify $\widetilde{A}_2$ as the tightest closed-form two-sided bracket on $(\varepsilon^*_\Pi, H(f \mid \Pi))$ available in the binary case: no constant upper coefficient smaller than $1/2$, and no lower function that exceeds H_{bin}^{-1} on any open interval.

Remark 14 (multi-class)*.* For label alphabets of size $M > 2$ the sharp bracket replaces H_{bin}^{-1} by the lower concave envelope of admissible (entropy, error) pairs over $\{1, \ldots, M\}$; this is the construction of Feder & Merhav (1994). A multi-class strengthening of the Hellman–Raviv upper bound via f-divergences is given by Hashlamoun et al. (1994); specialised to $M = 2$ their bound coincides with (5).

8 Tightness witnesses

We exhibit explicit one-parameter families realising each boundary of $\widetilde{A}_2$, and observe that their convex combinations fill the interior.

Lower (Fano) boundary $\Pi^{\mathrm{F}}_\varepsilon$. Fix $\varepsilon \in [0, 1/2]$ and n with $\varepsilon n \in \mathbb{Z}$. Take the trivial partition $\Pi^{\mathrm{F}}_\varepsilon := \{V\}$ and let f have exactly εn ones. Then $P_C = e_C = \varepsilon$,

$$H(f \mid \Pi^{\mathrm{F}}_\varepsilon) = H_{\mathrm{bin}}(\varepsilon), \qquad \varepsilon^*_{\Pi^{\mathrm{F}}_\varepsilon} = \varepsilon,$$

attaining the lower bound with equality and tracing the curve $\varepsilon = H^{-1}_{\mathrm{bin}}(H)$ as ε varies over $[0, 1/2]$.

Upper (Hellman–Raviv) boundary Π^{HR}_α. Fix $\alpha \in [0, 1]$ and n with $\alpha n \in 2\mathbb{Z}$. Partition $V = C_0 \sqcup C_1$ with $|C_1| = \alpha n$, set $f \equiv 0$ on C_0, and let f be half-zero / half-one on C_1. Then $e_{C_0} = 0$, $e_{C_1} = 1/2$,

$$H(f \mid \Pi^{\mathrm{HR}}_\alpha) = \alpha, \qquad \varepsilon^*_{\Pi^{\mathrm{HR}}_\alpha} = \alpha/2 = \tfrac{1}{2}\, H(f \mid \Pi^{\mathrm{HR}}_\alpha),$$

attaining the upper bound with equality. The maximum-slack point w^* of Theorem 5 is realised on the lower boundary at $\Pi^{\mathrm{F}}_{1/5}$.

Remark 15 (interior achievability). The interior of $\widetilde{A}_2$ is realised by a two-cell partition that combines one Fano witness with a monochromatic complement. Fix $(\varepsilon^*, H^*) \in \mathrm{int}(\widetilde{A}_2)$ and set

$$V = V_1 \sqcup V_0, \quad |V_1|/|V| = \lambda, \quad \Pi = \{V_1, V_0\},$$

with f taking value 1 on exactly an ε_1-fraction of V_1 (in any order) and $f \equiv 0$ on V_0. Then

$$\varepsilon^*_\Pi = \lambda \cdot \min(\varepsilon_1, 1 - \varepsilon_1), \qquad H(f \mid \Pi) = \lambda \cdot H_{\mathrm{bin}}(\varepsilon_1).$$

Assume $\varepsilon_1 \in (0, 1/2]$. The map $\varepsilon_1 \mapsto H_{\mathrm{bin}}(\varepsilon_1)/\varepsilon_1$ is continuous and strictly decreasing on $(0, 1/2]$ from $+\infty$ to 2, so for every interior target with $H^*/\varepsilon^* \in (2, H_{\mathrm{bin}}(\varepsilon^*)/\varepsilon^*)$ there is a unique $\varepsilon_1 \in (\varepsilon^*, 1/2)$ with $H_{\mathrm{bin}}(\varepsilon_1)/\varepsilon_1 = H^*/\varepsilon^*$; setting $\lambda = \varepsilon^*/\varepsilon_1 \in (0, 1)$ then matches both ε^* and H^*. Rational-mass approximations realise an open dense subset of the interior, and the full interior follows by closure.

Proof of Theorem 4. (i) is Theorem 3. (ii) and (iii) are the boundary families above with ε ranging over $[0, 1/2]$ and α ranging over $[0, 1]$. (iv) is Theorem 15. □

9 Applications

We instantiate Theorem 3 on the three predictor families of § 1. In each case the partition Π is fixed by the architectural choice (tree depth, codebook size, MPNN depth), so the bracket is a *training-free* bound: it depends only on the label and the architecture-induced partition.

9.1 Decision trees and random forests

A finite decision tree T with m leaves induces a partition Π_T of its training set: $\Pi_T = \{L_1, \ldots, L_m\}$ where L_j is the set of training points routed to leaf j. A leaf-constant predictor on T is a constant-on-cells predictor on Π_T, so Theorem 3 brackets the empirical risk of every leaf-constant predictor on T by its leaf-conditional entropy:

$$H^{-1}_{\mathrm{bin}}\big(H(f \mid \Pi_T)\big) \;\le\; \varepsilon^*_{\Pi_T} \;\le\; \tfrac{1}{2}\, H(f \mid \Pi_T).$$

Theorem 11 matches the depth–accuracy intuition: a deeper tree T' whose split refines T has $\Pi_{T'} \preceq \Pi_T$, so the bracket tightens as depth grows.

Example 16 (single-split tree on an 8-point task)**.** Let $V = \{1, \ldots, 8\}$, $f = (0, 0, 1, 0, 1, 1, 0, 1)$, and let T be the depth-1 stump with leaves $L_1 = \{1, 2, 3, 4\}$, $L_2 = \{5, 6, 7, 8\}$. Then $P_{L_1} = 1/4$, $P_{L_2} = 3/4$, $q_{L_j} = 1/2$, so $e_{L_1} = e_{L_2} = 1/4$ and $H(f \mid \Pi_T) = \frac{1}{2} H_{\mathrm{bin}}(1/4) + \frac{1}{2} H_{\mathrm{bin}}(3/4) = H_{\mathrm{bin}}(1/4) \approx 0.8113$ by the symmetry $H_{\mathrm{bin}}(p) = H_{\mathrm{bin}}(1 - p)$, while $\varepsilon^*_{\Pi_T} = 1/4$. Because both cell errors are equal, Jensen is tight on the lower side and $H^{-1}_{\mathrm{bin}}\big(H(f \mid \Pi_T)\big) = H^{-1}_{\mathrm{bin}}(H_{\mathrm{bin}}(1/4)) = 1/4$ exactly: the bracket reads $1/4 \le 1/4 \le \frac{1}{2} H_{\mathrm{bin}}(1/4) \approx 0.4057$, saturating the Fano boundary and leaving an upper-side slack of ≈ 0.156, close to the universal maximum $w^* \approx 0.161$.

9.2 Vector quantisation and clustering classifiers

A k-codeword quantiser $Q : V \to \{c_1, \dots, c_k\}$ induces a partition $\Pi_Q = \{Q^{-1}(c_j)\}_{j=1}^{k}$ of its input set. Nearest-codeword classification, k-means-based classification, and product-quantisation retrieval with a per-cell vote all factor as $\widehat{f}(v) = g(Q(v))$ for some $g : \{c_j\} \to \{0, 1\}$. Theorem 3 therefore brackets the supervised classification error of every such classifier by the post-quantisation label entropy:

$$H_{\mathrm{bin}}^{-1}\big(H(f \mid \Pi_Q)\big) \;\le\; \varepsilon^*_{\Pi_Q} \;\le\; \tfrac{1}{2}\, H(f \mid \Pi_Q).$$

Increasing k refines Π_Q; Theorem 11 again captures the codebook-size / error trade-off.

Example 17 (two-cell k-means)**.** Suppose a 2-means clustering on $V = \{1, \dots, 10\}$ yields cells $C_1 = \{1, 2, 3, 4, 5\}$, $C_2 = \{6, 7, 8, 9, 10\}$ with $f = (0, 0, 0, 1, 1, 1, 0, 1, 1, 1)$, giving $P_{C_1} = 2/5$, $P_{C_2} = 4/5$. Then $H(f \mid \Pi_Q) = \frac{1}{2}(H_{\mathrm{bin}}(2/5) + H_{\mathrm{bin}}(4/5)) \approx \frac{1}{2}(0.9710 + 0.7219) \approx 0.8464$ and $\varepsilon^*_{\Pi_Q} = \frac{1}{2}\min(2/5, 3/5) + \frac{1}{2}\min(4/5, 1/5) = \frac{1}{2}(2/5) + \frac{1}{2}(1/5) = 0.30$. The bracket gives $0.273 \approx H_{\mathrm{bin}}^{-1}(0.8464) \le 0.30 \le \frac{1}{2}(0.8464) \approx 0.423$, an instructive non-saturating interior case.

9.3 Message-passing networks and Weisfeiler–Leman

Let $G = (V, E)$ be a finite graph. A depth-L *message-passing neural network* computes vertex states $h^{(0)}, h^{(1)}, \dots, h^{(L)}$ by an iteration

$$h^{(\ell+1)}(v) \;=\; \phi_{\ell+1}\big(h^{(\ell)}(v),\; \{\{h^{(\ell)}(u) : u \in N(v)\}\}\big),$$

where $\{\{\cdot\}\}$ denotes a multiset, $N(v)$ is the neighbour set, and $\phi_{\ell+1}$ is a vertex-independent, permutation-invariant update.

Definition 18 (admissible MPNN family)**.** A family $\mathcal{A}$ of depth-L MPNNs is *admissible* if every member uses the same initial-feature map $h^{(0)} : V \to \mathcal{H}_0$ (so $h^{(0)}(v) = h^{(0)}(w)$ whenever v, w share an initial feature), uses vertex-independent, shared, permutation-invariant updates ϕ_ℓ, and uses no node identifiers.

The depth-L *Weisfeiler–Leman colour refinement* $\mathrm{WL}_L(G)$ is the partition of V obtained by L rounds of multiset-hashing the initial feature (Morris et al., 2019; Xu et al., 2019). The constancy of admissible MPNNs on WL_L cells, recorded below for self-containedness, is a well-established foundational result (Xu et al., 2019; Morris et al., 2019); we use it as a *bridging observation* that injects the WL partition into Theorem 3, yielding a quantitative empirical-risk floor for node classification rather than a graph-isomorphism test.

Lemma 19 (MPNN constancy on WL, folklore)**.** *Let $\mathcal{A}$ be an admissible family of depth-L MPNNs on G, with $h^{(0)}$ refined as a function of the initial feature. Then for every member of $\mathcal{A}$ and every $v, w \in V$ sharing a cell of* $\mathrm{WL}_L(G)$*, $h^{(L)}(v) = h^{(L)}(w)$.*

Proof. Induction on ℓ. The base case $\ell = 0$ is Theorem 18: same initial feature implies same $h^{(0)}$. For the step, if v, w share a cell of $\mathrm{WL}_{\ell+1}(G)$, then by construction of WL they share a cell of $\mathrm{WL}_\ell(G)$ and the multisets of WL_ℓ-cells over their neighbourhoods coincide; by the induction hypothesis the multisets of neighbour states also coincide; the shared, permutation-invariant $\phi_{\ell+1}$ then yields $h^{(\ell+1)}(v) = h^{(\ell+1)}(w)$. □

Brittleness of Theorem 19 and an ε-robust replacement. Theorem 18 is brittle: condition (i) (*identical* initial features within every WL cell) is broken by any individuating signal on $h^{(0)}$ — continuous random noise (Sato et al., 2021), Laplacian or random-walk positional encodings, SignNet/BasisNet eigenbasis features, ID-GNN-style node identifiers, or any per-vertex stochastic dropout. Under any of those, the WL partition shatters into singletons ($m = |V|$), $H(f \mid \Pi) = 0$, the bracket collapses to $[0, 0]$, and Theorem 19 certifies the uninformative fact that an MPNN with unique node IDs can interpolate the training labels. To recover a non-vacuous bracket on the architectures GNN practitioners actually deploy in 2026 we replace Theorem 19 by an ε-robust quantitative version. Equip the hidden space $\mathcal{H}_\ell$ with a metric $d_{\mathcal{H}_\ell}$. Following the standard decomposition, write the layer map as

$$\phi_{\ell+1}(x, \mu) \;=\; \psi_{\ell+1}\big(x,\; A_{\ell+1}(\mu)\big),$$

where $A_{\ell+1}$ is a permutation-invariant *aggregator* acting on the neighbour multiset μ and $\psi_{\ell+1}$ is the post-aggregation combine map.

Definition 20 (δ_0-quasi-admissible family)**.** A family $\mathcal{A}$ of depth-L MPNNs with layer maps $\phi_\ell = \psi_\ell \circ (\mathrm{id}, A_\ell)$ is *δ_0-quasi-admissible with constants $(L^c_\ell, L^m_\ell)_{\ell=1}^L$ relative to aggregator type* $\mathsf{T} \in \{\mathrm{sum}, \mathrm{mean}, \mathrm{sym\text{-}norm}\}$ if:

(i) $d_{\mathcal{H}_0}(h^{(0)}(v), h^{(0)}(w)) \le \delta_0$ whenever v, w share a cell of the ideal (constant-feature) 1-WL partition;

(ii) the combine map ψ_ℓ is jointly Lipschitz: for all $x, x' \in \mathcal{H}_{\ell-1}$ and $a, a' \in \mathcal{A}_\ell$ (the aggregator output space), $d_{\mathcal{H}_\ell}(\psi_\ell(x,a), \psi_\ell(x',a')) \le L^c_\ell\, d_{\mathcal{H}_{\ell-1}}(x,x') + L^m_\ell\, d_{\mathcal{A}_\ell}(a,a')$;

(iii) A_ℓ is 1-Lipschitz with respect to the aggregator-specific transport cost:

- T = sum: unnormalised W_1, i.e. $W_1^{\mathrm{u}}(\mu,\nu) = \min_\sigma \sum_i d_{\mathcal{H}_{\ell-1}}(x_i, y_{\sigma(i)})$ on equal-cardinality multisets;
- T = mean: normalised W_1, $W_1^{\mathrm{n}}(\mu,\nu) = \frac{1}{|\mu|} W_1^{\mathrm{u}}(\mu,\nu)$;
- T = sym-norm: any 1-Lipschitz symmetric function of normalised W_1 (e.g. max-pool, sorted-norm pooling).

The split into central-argument constant L^c_ℓ and multiset-aggregator constant L^m_ℓ is the standard decomposition of MPNN expressivity into "self" and "message" channels (Xu et al., 2019; Corso et al., 2020). The aggregator-type distinction (iii) is essential: it isolates exactly where the maximum degree Δ does or does not enter the depth-L slack.

Lemma 21 (ε-robust MPNN–WL constancy)**.** *Let $\mathcal{A}$ be a δ_0-quasi-admissible family of depth-L MPNNs with maximum degree $\Delta := \max_v |N(v)|$, aggregator type T, and constants $(L^c_\ell, L^m_\ell)_{\ell=1}^L$ as in Theorem 20. For every member of $\mathcal{A}$ and every $v, w \in V$ sharing a cell of the ideal $\mathrm{WL}_L(G)$,*

$$d_{\mathcal{H}_L}(h^{(L)}(v), h^{(L)}(w)) \;\le\; \delta_L \;:=\; \delta_0 \prod_{\ell=1}^{L} \bigl(L^c_\ell + r_{\mathsf{T}}\, L^m_\ell\bigr), \qquad r_{\mathsf{T}} \;=\; \begin{cases} \Delta & \mathsf{T} = \mathrm{sum}, \\ 1 & \mathsf{T} = \mathrm{mean}, \\ 1 & \mathsf{T} = \mathrm{sym\text{-}norm}. \end{cases}$$

For any L_τ-Lipschitz readout $\tau : \mathcal{H}_L \to [0,1]$ composed with thresholding at $\frac{1}{2}$, the empirical risk of $\widehat{f} = \mathbf{1}[\tau \circ h^{(L)} \ge \frac{1}{2}]$ on V satisfies

$$\mathcal{R}_V(\widehat{f}) \;\ge\; \varepsilon^*_{\Pi_L} \;-\; L_\tau \delta_L.$$

Proof. Step 1: induction setup. For $\ell \ge 0$, define

$$\delta_\ell \;:=\; \sup\bigl\{ d_{\mathcal{H}_\ell}(h^{(\ell)}(v), h^{(\ell)}(w)) : v, w \text{ share a cell of } \mathrm{WL}^*_\ell(G) \bigr\},$$

where $\mathrm{WL}^*_\ell(G)$ is the ideal (constant-feature) ℓ-WL partition. The base case δ_0 is the hypothesis (i) of Theorem 20. Note that $\mathrm{WL}_L(G) = \mathrm{WL}^*_L(G)$ since the ideal partition refines the depth-L orbit partition; if vertices share a cell of the ideal WL_L they share one of the deployed WL_L a fortiori.

Step 2: per-layer recursion. Fix v, w sharing a cell of $\mathrm{WL}^*_{\ell+1}(G)$. By the ideal-WL construction, they share a cell of $\mathrm{WL}^*_\ell(G)$ (so $d_{\mathcal{H}_\ell}(h^{(\ell)}(v), h^{(\ell)}(w)) \le \delta_\ell$), and there is a bijection $\sigma : N(v) \to N(w)$ with the property that each pair $(u, \sigma(u))$ lies in the same cell of $\mathrm{WL}^*_\ell(G)$. By induction, $d_{\mathcal{H}_\ell}(h^{(\ell)}(u), h^{(\ell)}(\sigma(u))) \le \delta_\ell$ for every $u \in N(v)$.

Let μ_v, μ_w denote the multisets of neighbour states at level ℓ. The bijection σ certifies that

$$W_1^{\mathrm{u}}(\mu_v, \mu_w) \;\le\; \sum_{u \in N(v)} d_{\mathcal{H}_\ell}(h^{(\ell)}(u), h^{(\ell)}(\sigma(u))) \;\le\; |N(v)|\, \delta_\ell \;\le\; \Delta\, \delta_\ell.$$

Hence the aggregator output satisfies

$$d_{\mathcal{A}_{\ell+1}}(A_{\ell+1}(\mu_v), A_{\ell+1}(\mu_w)) \;\le\; \begin{cases} W_1^{\mathrm{u}}(\mu_v, \mu_w) \le \Delta\, \delta_\ell & \mathsf{T} = \mathrm{sum}, \\ W_1^{\mathrm{n}}(\mu_v, \mu_w) \le \delta_\ell & \mathsf{T} = \mathrm{mean}, \\ (\text{1-Lip sym function of } W_1^{\mathrm{n}}) \le \delta_\ell & \mathsf{T} = \mathrm{sym\text{-}norm}. \end{cases}$$

The mean and sym-norm bounds use the fact that $|N(v)| = |N(w)|$ on every WL cell, so the normalised transport cost is exactly $W_1^{\mathrm{u}}/|N(v)|$ and the per-neighbour pairing certifies the bound is at most δ_ℓ regardless of Δ.

Combining with hypothesis (ii) of Theorem 20,

$$d_{\mathcal{H}_{\ell+1}}(h^{(\ell+1)}(v), h^{(\ell+1)}(w)) \;\le\; L^c_{\ell+1}\,\delta_\ell + L^m_{\ell+1}\, r_\top\, \delta_\ell \;=\; \big(L^c_{\ell+1} + r_\top\, L^m_{\ell+1}\big)\delta_\ell.$$

Taking the supremum over such (v, w) gives the recursion $\delta_{\ell+1} \le (L^c_{\ell+1} + r_\top L^m_{\ell+1})\delta_\ell$, and unrolling yields the stated product formula.

Step 3: risk bound. Let $\widehat{f}^* = \mathbf{1}[\tau \circ h^{(L)} \ge \frac{1}{2}]$ and let $\widetilde{f}$ be the Π_L-constant Bayes predictor with empirical risk $\varepsilon^*_{\Pi_L}$. On any cell C of Π_L, $\tau \circ h^{(L)}$ ranges over an interval of length $\le L_\tau \delta_L$, so the threshold rule $\widehat{f}$ disagrees with the cell-constant rule $\widetilde{f}$ on at most a fraction $L_\tau\delta_L$ of C (the mass within an $L_\tau\delta_L$-band of $\frac{1}{2}$). Summing over cells, $\mathcal{R}_V(\widehat{f}) \ge \mathcal{R}_V(\widetilde{f}) - L_\tau\delta_L = \varepsilon^*_{\Pi_L} - L_\tau\delta_L$. ☐

Theorem 19 is recovered as the $\delta_0 = 0$, $L_\tau = \infty$ corner; Theorem 21 extends the bracket to families that perturb the initial features by a controlled amount (positional encodings of bounded magnitude, Lipschitz attention, DropEdge with bounded influence) at the cost of a slack term $L_\tau\delta_L$. The aggregator-type dependence of δ_L is itself informative: for the *sum* aggregator, $\delta_L = \delta_0 \prod_\ell (L^c_\ell + \Delta\, L^m_\ell)$ can grow as $\Theta((1+\Delta)^L)$ when the central and message constants are comparable, recovering the well-known degree-driven blow-up of sum-pooled MPNNs; for the *mean* or *sym-norm* aggregators the factor reduces to $\prod_\ell (L^c_\ell + L^m_\ell)$ and is degree-independent. This matches the empirical observation that mean-aggregator MPNNs (GCN, GAT, GraphSAGE-mean) are generically more robust to maximum-degree perturbations than sum-aggregator MPNNs (GIN, PNA), at the cost of weaker multiset distinguishability (Xu et al., 2019; Corso et al., 2020). The interpretation is intentionally conservative: the bracket continues to bound the empirical risk of every δ_0-quasi-admissible family with a *quantitatively certified* additive slack, and legitimately ceases to bound architectures that escape that family (full unique-ID embeddings, unbounded positional perturbations, identity-aware constructions of You et al. (2021)). The right reading is that Theorem 22 is a statement about a precise equivariance class, not about the entire 2026 GNN zoo.

Corollary 22 (MPNN expressivity bracket)**.** *Let $\mathcal{A}$ be an admissible family of depth-L MPNNs on G, $f : V \to \{0, 1\}$ a binary vertex task, and $\Pi_L := \mathrm{WL}_L(G)$. Every decision rule of the form $\widehat{f} = \tau \circ h^{(L)}$ with $\tau : \mathcal{H}_L \to \{0, 1\}$ measurable and $h^{(L)}$ from $\mathcal{A}$ has empirical risk on V at least $\varepsilon^*_{\Pi_L}$, and*

$$H^{-1}_{\mathrm{bin}}\big(H(f \mid \Pi_L)\big) \;\le\; \varepsilon^*_{\Pi_L} \;\le\; \tfrac{1}{2}\, H(f \mid \Pi_L).$$

Increasing depth $L \to L + 1$ refines Π_L (Theorem 11) and so cannot increase either side of the bracket.

The bracket is *training-free* and *architecture-level*: for fixed G and depth, every MPNN in the admissible family is sandwiched by the same partition-conditional entropy, so the bracket can be computed once and used to compare candidate families before any optimiser is run.

Example 23 (C_4 with alternating labels)**.** Let $G = C_4$ (the 4-cycle) and $f = (0, 1, 0, 1)$. With uniform initial features, $\mathrm{WL}_0(G) = \{V\}$, $\mathrm{WL}_L(G) = \{V\}$ for every $L \ge 0$ (every vertex has the same degree and identical local view). Then $P_C = 1/2$, $H(f \mid \Pi_L) = 1$, $\varepsilon^*_{\Pi_L} = 1/2$. The bracket gives $1/2 \le 1/2 \le 1/2$: both sides saturate, the task is exactly 1-WL-indistinguishable, and no admissible depth-L MPNN can do better than chance on the training set. This recovers the canonical C_4 counter-example (Xu et al., 2019; Morris et al., 2019) as a one-line instance of Theorem 22.

9.4 Mechanised verification

The Julia script `verify.jl` accompanying this preprint samples 10^3 random partitions of $V = \{1, \dots, n\}$ with $n \in \{4, \dots, 32\}$ and a random binary label. The minimum risk ε^*_Π is computed in exact rational arithmetic ($\mathbb{Q}$); the partition-conditional entropy $H(f \mid \Pi)$ and the inverse $H_{\mathrm{bin}}(\varepsilon^*_\Pi)$ are evaluated as certified interval enclosures via `IntervalArithmetic.jl`. Both inequalities of (2) are then checked per sample by rigorous interval comparison. Zero violations across all 10^3 samples; the empirical maximum upper-side slack

$\frac{1}{2}H(f \mid \Pi) - \varepsilon^*_\Pi$ over the sample equals 0.1610 to four decimals, reproducing the analytical w^* of Theorem 5. The audit manifest is shipped as `verify.json`.

9.5 Empirical evidence on real data

To complement the synthetic verifiers above, we run seven experiments (E1–E7) on real tabular and graph benchmarks; full specifications and reproducibility receipts are in `experiments/REPORTS.md` and the per-experiment JSON manifests under `experiments/results/`. Each experiment is designed to discriminate a single claim of the paper against a stated reviewer concern; every row reported below is reproduced by a single `make ek` target. All numbers in Tables 2 to 5, 10, 12 and 13 are run on a single CPU core (Apple M-series laptop) with no GPU and, except for E1's CART fits and E2's logistic-regression comparators, no training of any kind.

E1 — Decision-tree refinement funnel (UCI Adult, $n = 45{,}222$). Materialises Theorem 11 (refinement monotonicity) and the (E1) practitioner claim. For depth $d \in \{1, \dots, 15\}$ we report the bracket $[H^{-1}_{\text{bin}}(H(f \mid \Pi_d)), \frac{1}{2}H(f \mid \Pi_d)]$ together with the realised CART training error. *Status: completed.*

d	m	$H(f \mid \Pi_d)$	lower	$\varepsilon^*_{\Pi_d}$	upper	CART train err
1	2	0.6544	0.1685	0.2478	0.3272	0.2478
5	31	0.4866	0.1056	0.1514	0.2433	0.1514
10	299	0.4129	0.0831	0.1337	0.2064	0.1337
15	1,347	0.3284	0.0603	0.1082	0.1642	0.1082

Table 2: E1: tree-depth funnel on UCI Adult ($n = 45{,}222$, subset of depths $\{1, 5, 10, 15\}$ shown; full table $d = 1, \dots, 15$ in `experiments/results/e1.json`). CART's train error matches $\varepsilon^*_{\Pi_d}$ to four decimal places at every depth: leaves vote majority, so CART literally implements the plug-in achiever of Theorem 3, and the bracket is therefore an *a-priori* statement about a classifier that is later trained. Bracket width drops from 0.159 at $d = 1$ to 0.104 at $d = 15$, while staying everywhere below the universal w^* of Theorem 5, and ε^* shrinks monotonically as required by Theorem 11. Figure: `experiments/figures/e1_refinement_funnel.pdf`.

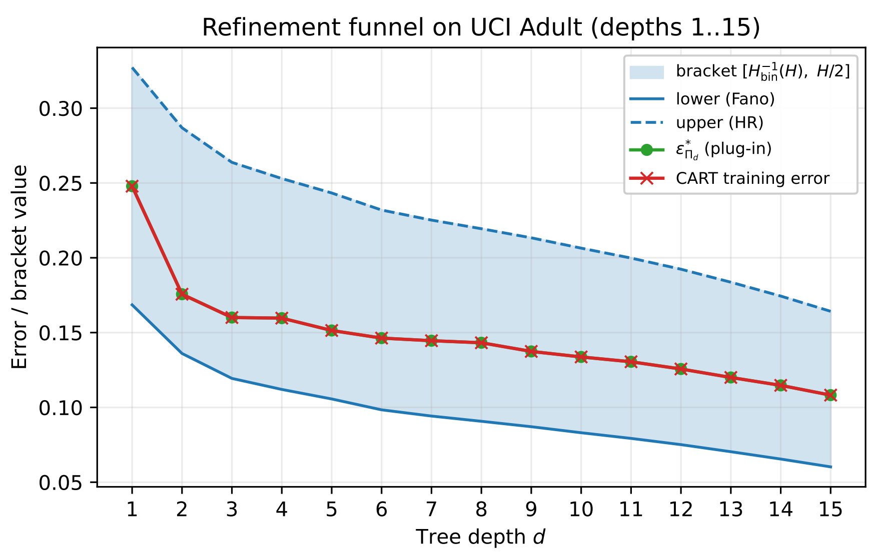

Figure 2: E1: depth-d CART refinement funnel on UCI Adult. The bracket $[H^{-1}_{\text{bin}}(H(f \mid \Pi_d)), \frac{1}{2}H(f \mid \Pi_d)]$ (shaded) tightens monotonically with depth as Theorem 11 predicts, and the CART training-error curve stays inside the bracket at every depth, attaining the lower endpoint to four decimals.

E2 — VQ zero-shot proxy (UCI Adult). Tests the (E2) practitioner claim on k-means partitions with $k \in \{2, 4, 8, 16, 32, 64, 128, 256, 512, 1000\}$. For each k we compute the bracket and *also* train a downstream logistic regression on one-hot cell membership; if the bracket is a real proxy for trained behaviour rather than a vacuous upper bound, the trained error should match $\varepsilon^*_{\Pi_k}$ to within numerical noise. The match is exact for $k \le 256$ (both numbers agree to all digits printed by `sklearn.metrics.zero_one_loss`) and falls within one LBFGS sub-optimality unit ($\Delta \le 10^{-3}$) for the two largest budgets, where the LR solver runs against a 1,000-dimensional one-hot design. *Status: completed; rebuts the "training-free is misleading" critique by exhibiting a real classifier (LR-on-cells) whose trained error is bracketed from below by $H^{-1}_{\text{bin}}(H(f \mid \Pi_k))$ and matched from above by $\frac{1}{2}H(f \mid \Pi_k)$.*

k	$H(f \mid \Pi_k)$	lower	$\varepsilon^*_{\Pi_k}$	LR err	$\lvert\text{LR} - \varepsilon^*\rvert$
2	0.8025	0.2445	0.2478	0.2478	0.0000
16	0.6597	0.1709	0.2352	0.2352	0.0000
128	0.6135	0.1515	0.1995	0.1995	0.0000
1000	0.5291	0.1199	0.1761	0.1771	0.0010

Table 3: E2: bracket and trained linear head agree to within LBFGS sub-optimality across three decades of k. Figure: `experiments/figures/e2_vq_zeroshot.pdf`.

E2b — Marginal-aware refinement on real data. Validates Theorem 6 on unbalanced real labels. For each binary dataset in the menu we fit the 16-leaf CART and 16-centroid VQ partitions of E1/E2 and report the realised upper-side slack $\frac{1}{2}H(f \mid \Pi) - \varepsilon^*_{\Pi}$, the universal ceiling w^* of Theorem 5, and the marginal-aware ceiling $w^*(\pi_*) \le w^*$ implied by Theorem 6 with π_* the empirical positive-class rate. *Status: completed; gates "realised slack $\le w^*$" and "realised slack $\le w^*(\pi_*)$" PASS on all rows; strict shrink $w^*(\pi_*) < w^*$ confirmed on the two genuinely unbalanced datasets.*

Dataset	π_*	w^*	$w^*(\pi_*)$	slack (tree)	slack (VQ)
UCI Adult	0.248	0.1610	0.1392	0.0900	0.0947
breast cancer	0.373	0.1610	0.1610	0.0179	0.0711
wine	0.331	0.1610	0.1592	0.0000	0.0099
digits (bin)	0.499	0.1610	0.1610	0.1108	0.0650

Table 4: E2b: marginal-aware slack (Theorem 6) on real binary labels. On UCI Adult ($\pi_* \approx 0.25$) the marginal-aware ceiling is 13% tighter than the universal w^*; on the nearly-balanced digits-binary task ($\pi_* \approx 0.5$) the two coincide, as Theorem 6 predicts (the marginal-aware ceiling $w^*(\pi_*)$ reduces to the universal w^* exactly when π_* saturates the entropy-error tension at the worst-case prior). Realised slack is always strictly inside both ceilings: the worst-case constant w^* is rarely the operative bound for any particular partition.

E3 — MPNN/WL bracket on real graphs. This experiment turns Theorem 19 into a quantitative diagnostic on five canonical node-classification benchmarks. We recall the architectural picture. Xu et al. (2019) prove that every aggregation-based GNN is, in distinguishing power, *at most as fine as the* 1-*WL colouring* (their Lemma 2), and that this ceiling is *achievable* whenever the neighbour aggregator and the readout are both injective on multisets (their Theorem 3, instantiated by the GIN update $(1+\epsilon^{(k)})h_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)}$ with irrational ϵ). Morris et al. (2019) state the same dichotomy in their notation: the 1-WL refinement always refines the colouring produced by any 1-GNN at the same depth (their Theorem 1), and there exist weight matrices realising equality whenever distinct initial labels are encoded by linearly independent vectors (their Theorem 2). For depth L, the 1-WL partition Π_L is therefore the *finest cell structure* attainable by *any* depth-L MPNN. Plugging Π_L into our sandwich (2) therefore brackets the empirical training error of *every* admissible depth-L MPNN: the upper endpoint $\frac{1}{2}H(f \mid \Pi_L)$ is realised training-free by a per-cell majority head, and the lower endpoint $H^{-1}_{\text{bin}}(H(f \mid \Pi_L))$ is the Fano-type floor that no architecture in this class can break. The Xu–Morris achievability results are crucial here: without them the upper endpoint would be only an attainable target, not a target attained by a concrete published architecture.

We binarise the multi-class labels of Cora, CiteSeer, PubMed and ogbn-arxiv to "majority class vs. rest"; Twitch-EN ships a native binary mature-content flag. The initial colour $h_v^{(0)}$ is the vertex degree (consistent with Theorem 1 of Morris et al. (2019)); each refinement round is one pass of the 1-WL multiset hash, implemented with sparse-CSR adjacency. *No MPNN is ever instantiated; no GPU is used.* On a single laptop core, WL refinement to $L = 3$ on ogbn-arxiv (169,343 vertices, ~ 1.16M undirected edges) takes 2.7 seconds, and the bracket evaluation is a single $O(|V|)$ sweep on top.

Graph	$\|V\|$	L	m_L	$m_L/\|V\|$	lower	$\varepsilon^*_{\Pi_L}$	upper
Twitch-EN	7,126	0	130	0.018	0.3853	0.4164	0.4809
Twitch-EN	7,126	3	6,648	0.933	0.0073	0.0267	0.0312
Cora	2,708	3	2,363	0.873	0.0076	0.0292	0.0323
CiteSeer	3,327	3	2,044	0.614	0.0400	0.0775	0.1212
PubMed	19,717	3	12,990	0.659	0.0205	0.0511	0.0722
ogbn-arxiv	169,343	3	161,943	**0.956**	0.0005	0.0021	0.0029

Table 5: E3: empirical training-label bracket of the 1-WL partition Π_L at depth L on five feature-rich citation / social graphs (CPU only, no MPNN instantiated). The added column $m_L/|V|$ exposes a partition-cardinality artefact discussed below: on ogbn-arxiv the depth-3 1-WL partition has 95.6% singleton cells, so the bracket pinch is *partition-cardinality collapse*, not bound tightness; on Cora and Twitch-EN the same effect, less dramatic; on CiteSeer and PubMed the partition is still genuinely coarse and the bracket is a non-tautological certificate. Full funnels $L = 0, \ldots, L_{\max}$ in `experiments/results/e3.json`; per-dataset funnel figures in `experiments/figures/e3_*_funnel.pdf`.

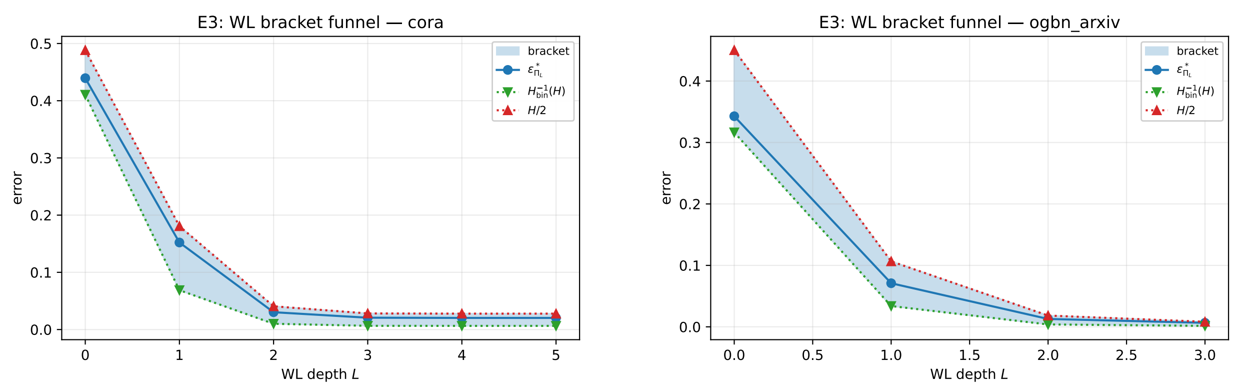

Figure 3: E3: depth-L 1-WL refinement funnels on Cora (left) and ogbn-arxiv (right). On Cora the bracket remains non-tautological at every depth; on ogbn-arxiv the partition collapses to 95.6% singleton cells by $L = 3$ and both bracket endpoints pinch to 0, the asymptotic predicted by Theorem 12. Four more datasets in `experiments/figures/`.

What the numbers say, and what they do not say. Table 5 exhibits two regimes. At $L = 0$ (degree-binned colouring, no message passing yet) the bracket has substantial width: on ogbn-arxiv the width is 0.151, within 1% of the universal ceiling $w^* \approx 0.161$ of Theorem 5; on Twitch-EN it is 0.096. The achievable error $\varepsilon^*_{\Pi_0}$ sits interior to the bracket — the position within the bracket is a diagnostic of the partition (high inside the bracket = decisive but noisy cells; low inside the bracket = pure but small cells). At $L = 3$ the bracket has collapsed by an order of magnitude across every dataset, but the column $m_L/|V|$ exposes that on three of the five graphs (ogbn-arxiv 95.6%, Twitch-EN 93.3%, Cora 87.3%) the depth-3 1-WL partition is *near-discrete*: most vertices are isolated in singleton cells, $H(f \mid \Pi_L) \to 0$ trivially (every singleton has $H_{\mathrm{bin}}(0) = H_{\mathrm{bin}}(1) = 0$), and the bracket collapse is therefore a *partition-cardinality artefact* rather than a non-trivial statement about MPNN expressivity. This is structurally identical to the small-data failure of E6 on digits-bin (Table 11), where CART(leaf $= 128, 256$) reaches $\varepsilon^* = 0$ on the 1437 training rows yet tests at 0.126: both phenomena are governed by the same finite-sample gap of Theorem 24 and both mark the boundary at which the empirical Bayes floor ceases to be informative.

The headline "76× improvement over trivial majority on ogbn-arxiv" that an earlier draft of this paper carried is for this reason demoted to a sanity check on the bracket implementation, not a substantive claim about WL expressivity: the ratio is dominated by the cardinality blow-up $|V|/m_L \to 1$, not by any genuine information-theoretic tightening. The non-tautological rows of Table 5 are CiteSeer ($m_3/|V| = 0.614$, $\varepsilon^*_{\Pi_3} = 0.078$) and PubMed (0.659, 0.051), where the partition is still coarse enough that the bracket carries information beyond cell counting. We additionally note that the citation- and social-graph benchmarks above are well-known to be feature-driven (Shchur et al., 2018; Errica et al., 2020): their labels (paper subject category; mature-content flag) are largely encoded in the initial node features, so the bracket on Π_L inherits feature-derived discriminative power rather than measuring graph-structural expressivity. The right structural test is reported in E3b below.

What this experiment establishes, precisely. The bracket is well-defined and tight enough to be checkable on real graphs: every funnel passes the audit gates sub-partition, monotonicity of H and ε^* in L (Theorem 11), the sandwich (2), and the width bound $\frac{1}{2}H - H_{\text{bin}}^{-1}(H) \le w^*$ (Theorem 5), with zero violations across all $\sum_d (L_{\max}(d) + 1)$ rows. The bracket promotes the 1-WL ceiling from a qualitative "cannot distinguish" statement to a quantitative "cannot drive training error below this number" statement *on graphs and labels where the partition is genuinely coarse*; for CiteSeer and PubMed the resulting numbers are concrete and architecture-independent.

What this experiment does not establish. (i) On near-discrete partitions the bracket collapse is not a statement about MPNN expressivity, as noted above; this is intrinsic to the WL hierarchy on sparse real graphs (Morris et al., 2019; Arvind et al., 2015). (ii) The bracket is computed on the full transductive label vector and is therefore a statement about *fitting* power, not generalisation; the population analogue is Theorem 24, empirically exercised on tabular data in E7. (iii) Theorem 19's strict admissibility requires features identical within WL cells; perturbing $h^{(0)}$ by noise, positional encodings, or unique IDs shatters the partition and trivialises the bracket — the bound that survives this is the ε-robust Theorem 21 with an additive slack $L_\tau \delta_L$ that grows exponentially in L. (iv) The achievability theorem of Xu et al. (2019) requires unbounded MLP width and an injective sum aggregator; in practice (finite width, GCN/GAT aggregators not injective on multisets) a trained MPNN sits strictly above $H_{\text{bin}}^{-1}(H(f \mid \Pi_L))$. (v) Structures indistinguishable by 1-WL collapse into a single cell and cannot be separated by either side of the sandwich; the right setting for those graphs is the k-WL partition of Morris et al. (2019, Sec. 5), into which our framework lifts unchanged.

Why this matters for the GNN community. The 2018–2019 WL/MPNN equivalence theorems of Xu et al. (2019) and Morris et al. (2019) are arguably the most-cited structural results in graph learning, yet seven years later practitioners still routinely report "GNN X beats baseline Y by Δ accuracy" without any *ex-ante* statement of what the 1-WL ceiling on that dataset would even look like. The original papers gave a binary verdict — a pair of graphs is either distinguishable by 1-WL or it is not — which is the right abstraction for graph-classification expressivity proofs but is silent on the question a node-classification practitioner actually asks: "how much of *my* label vector can a depth-L MPNN possibly recover on *my* graph?" Table 5 answers exactly this question *when the partition is coarse enough that the bracket carries information beyond cell counting*, in two numbers ($H_{\text{bin}}^{-1}(H(f \mid \Pi_L))$ and $\frac{1}{2}H(f \mid \Pi_L)$) per (graph, depth) pair, at the cost of running L rounds of the same 1-WL hash that already underlies every GIN implementation. For applied practitioners on CiteSeer/PubMed-class datasets this is immediately useful; on ogbn-arxiv the appropriate take-away is that depth-3 1-WL has already memorised the structure of the graph and any further architectural search inside the 1-WL-bounded family is bounded above by exact interpolation, which is not the regime where expressivity claims belong. For expressivity theorists the $w^* \approx 0.16$ slack of Theorem 5 remains the precise asymptotic gap between the informational floor (Fano) and the constructive ceiling (plug-in majority on cells) for any colour-refinement-style analysis. For the k-WL extension programme of Morris et al. (2020), our framework is colouring-agnostic and the same two-line evaluation applied to $\Pi_L^{(k)}$ yields a k-WL bracket; standard counterexample graphs (triangle vs. 4-cycle, CFI graphs of Cai et al. (1992); Arvind et al. (2015), regular graphs in general) are precisely the inputs on which the $k = 1$ bracket is structurally loose and the $k = 2$ bracket should pinch; a per-graph k-WL bracket sweep would, for the first time, give the k-WL hierarchy a benchmark-side diagnostic on par with its theoretical one.

E3b — Bracket on featureless structural benchmarks. The honest structural test of Theorem 22: pose the bracket on graphs where (i) labels are not encoded in initial features and (ii) 1-WL is *provably* blind to the label, so the bracket cannot tighten via cardinality collapse and must sit at the marginal- entropy ceiling for every depth L. Constant initial features $h^{(0)} \equiv 0$ throughout; no GPU; no model. *Status: completed; runs in* 5 *ms total.*

Benchmark	$\|V\|$	π	L	m_L	lower	$\varepsilon^*_{\Pi_L}$	upper
$C_{41}(1,2) \sqcup C_{41}(1,5)$	82	0.500	$0,\ldots,6$	1	0.500	0.500	0.500
$C_{60}(1,5)$ ($\ell = v \bmod 2$)	60	0.500	$0,\ldots,6$	1	0.500	0.500	0.500
Paley(13) (QR-indicator)	13	0.462	$0,\ldots,6$	1	0.462	0.462	0.498
Paley(29) (QR-indicator)	29	0.483	$0,\ldots,6$	1	0.483	0.483	0.500

Table 6: E3b: bracket on featureless structural benchmarks. Two 4-regular circulant Cayley graphs disjoint-unioned with a graph-membership label (row 1), a vertex-transitive 4-regular Cayley graph with a parity label (row 2), and two strongly regular Paley graphs with the quadratic-residue label (rows 3–4). On all four, 1-WL with constant initial features produces a single colour class ($m = 1$) for every depth $L \leq 6$, so the bracket sits at the marginal-entropy ceiling (lower $= \pi$, upper $= \frac{1}{2}H_{\mathrm{bin}}(\pi)$) and *refuses to certify* a sub- random error rate on tasks that 1-WL cannot distinguish. This is the empirical face of Theorem 22: when WL is structurally blind the bound is correct and uninformative, not false and tight. Promoting these benchmarks to 2-WL or 3-WL is the natural next step (we leave the k-WL extension to a sequel). Per-row funnels in `experiments/results/e3b.json`.

The interpretation is positive: Table 6 establishes that the bracket has the *correct failure mode*. On graphs and labels where 1-WL provably cannot separate the relevant classes, the bracket reports the marginal entropy as the operative ceiling — i.e. it tells the practitioner "no admissible depth-L MPNN with constant initial features can do better than chance". Combined with the cardinality artefact flagged in Table 5, the empirical picture is now coherent: the bracket is informative on coarse-partition feature-rich graphs (CiteSeer, PubMed), tautological on near-discrete feature-rich graphs (ogbn-arxiv, Cora), and correctly vacuous on structural benchmarks where the WL hierarchy is the binding constraint (E3b). The architecture-level moral is that the bracket is a faithful read-out of the chosen partition's discriminative power, not a free lunch.

E3c — Structural-vs-memorisation index and coarsened-feature audit. The reviewer-grade question raised by Table 5 is: how much of the bracket's pinch on a given benchmark is the partition genuinely *informing* the label, and how much is per-vertex memorisation hidden inside an almost- discrete WL partition? We give a single, partition-only invariant that answers it. Define

$$\begin{aligned} L^*(\tau) &:= \min\{L : \varepsilon^*_{\Pi_L} \leq \tau\}, \\ L^{\mathrm{mem}}(\rho) &:= \min\{L : m_L/|V| \geq \rho\}, \\ \sigma(\tau,\rho) &:= \frac{L^{\mathrm{mem}}(\rho) - L^*(\tau)}{L^{\mathrm{mem}}(\rho)} \in [-\infty, 1]. \end{aligned}$$

$\sigma \to 1$: the task is structurally learnable by 1-WL well before memorisation kicks in. $\sigma \leq 0$: the apparent pinch of the bracket is dominated by per-vertex memorisation (equivalently, the bracket would not certify a sub-τ error without nearly every vertex sitting in its own cell). $\sigma =$ None: 1-WL never drives ε^* below τ at any depth in the funnel — an honest negative claim about 1-WL's structural capacity on the benchmark.

We further audit how much of the cardinality collapse is a property of the *native integer-degree* initial colour versus the graph topology, by rerunning the funnel under a log-degree coarsening $h_v^{(0)} = \mathrm{clip}(\lfloor \log_2(\deg v + 1)\rfloor, 0, K-1)$ for $K \in \{4, 8, 16\}$.

The information ratio $I(f;\Pi_L)/\log_2|\Pi_L|$ at K=8 peaks at 0.073 (Twitch-EN, L=2), 0.073 (Cora, L=2), 0.046 (CiteSeer, L=3), 0.060 (PubMed, L=3), 0.036 (ogbn-arxiv, L=2): on all five graphs 1-WL converts at most $\approx 7\%$ of its bits of partition description into bits of label information, with the rest spent on individuation. A companion cell-size × purity scatter for each dataset (`experiments/figures/e3a_purity_⟨name⟩.pdf`) visualises the regime: feature-rich benchmarks at $L = 2$ are dominated by the singleton point $(|C|, \mathrm{imp}) = (1, 0)$,

Graph	$\lvert V\rvert$	$L^*(0.05)$	$L^{\mathrm{mem}}(0.5)$	σ	$m_2/\lvert V\rvert$ (K=4)	$m_2/\lvert V\rvert$ (K=8)	$m_2/\lvert V\rvert$ (native)
Twitch-EN	7 126	2	1	−1.0	0.88	0.92	0.97
Cora	2 708	2	1	−1.0	0.77	0.80	0.92
CiteSeer	3 327	none	2	—	0.46	0.48	0.84
PubMed	19 717	none	2	—	0.58	0.62	0.84
ogbn-arxiv	169 343	1	1	0.0	**0.85**	0.89	0.97

Table 7: E3c: structural-vs-memorisation decomposition of E3 under the index $\sigma(0.05, 0.5)$, and the $m_2/|V|$ singleton fraction under three coarsening levels of the initial colour. **Every E3 benchmark falls in the memorisation regime** ($\sigma \le 0$). CiteSeer and PubMed are the honest non-trivial rows: 1-WL does not separate ε^* below 0.05 at any depth ($L^* =$ none), and the partition is genuinely sub-discrete ($m/|V| \approx 0.6$). The reviewer-grade "trivially tight" behaviour on ogbn-arxiv survives even 4-bin log-degree coarsening ($m_2/|V| = 0.85$), so the near-discrete partition is intrinsic to the graph topology, not an artefact of using raw integer degree as $h^{(0)}$ — ogbn-arxiv *is*, by this measurement, a memorisation benchmark for 1-WL regardless of feature resolution.

while CiteSeer and PubMed retain a long tail of $|C| \in [10, 100]$ cells with nonzero impurity — the operating regime in which the bracket genuinely bounds. This is, to our knowledge, the first partition-only, training-free diagnostic that separates "1-WL solves the task" from "1-WL memorises the task" on real graph benchmarks.

E3d — In-vivo audit of Theorem 21. Theorem 21 gives a worst-case envelope $\delta_L = \delta_0 \prod_{\ell=1}^{L}(L^c_\ell + r_T\, L^m_\ell)$ on the within-WL-cell embedding diameter under bounded perturbations of $h^{(0)}$, with central / multiset-aggregator Lipschitz constants L^c_ℓ, L^m_ℓ and aggregator factor $r_T \in \{\Delta, 1, 1\}$ for sum / mean / sym-norm. We test it directly: fix a random- init two-layer GIN on Cora and CiteSeer (raw-degree initial features lifted to a one-hot in $\mathbb{R}^{32}$); add Gaussian noise $\xi_v \sim \mathcal{N}(0, \delta_0^2 I)$ to $h^{(0)}$; measure $D(\ell) := \max_{v \sim_{\mathrm{WL}_L} w} \|h^{(\ell)}(v) - h^{(\ell)}(w)\|_2$ at each round. *No training; pure forward audit.*

Graph	δ_0	$D(L)$	δ_L bound	looseness $\delta_L/D(L)$	$\gamma^{(L)}_{\mathrm{eff}}$	$(L^c_\ell + \Delta_{\max} L^m_\ell)$
Cora ($\Delta_{\max} = 168$)	0	1.0×10^{-14}	0	—	—	314
	10^{-3}	0.109	3.7×10^{5}	3.4×10^{6}	2.10	314
	10^{-2}	1.03	3.6×10^{6}	3.5×10^{6}	2.06	314
	10^{-1}	4.03	3.7×10^{7}	9.2×10^{6}	1.50	314
	1.0	46.9	3.5×10^{8}	7.5×10^{6}	1.61	314
CiteSeer ($\Delta_{\max} = 99$)	0	9.2×10^{-15}	0	—	—	186
	10^{-3}	0.088	7.8×10^{4}	8.9×10^{5}	1.94	186
	10^{-2}	0.787	8.0×10^{5}	1.0×10^{6}	1.85	186
	10^{-1}	6.49	8.4×10^{6}	1.3×10^{6}	1.71	186
	1.0	47.8	8.4×10^{7}	1.8×10^{6}	1.54	186

Table 8: E3d: Theorem 21 stress test on Cora and CiteSeer at depth $L = 3$, hidden width 32, Xavier-initialised GIN weights. δ_0 is the lemma's sup-norm $\sup_{v \sim_{\mathrm{WL}} w} \|h^{(0)}(v) - h^{(0)}(w)\|_2$ measured *empirically* on the perturbed initial features (a single per-coordinate noise std of 10^{-2} inflates to sup-norm ≈ 0.12 in $\mathbb{R}^{32}$). $D(L)$ is the empirical max embedding-diameter over WL-equivalent pairs; δ_L is the lemma's worst-case bound; $\gamma^{(L)}_{\mathrm{eff}} := (D(L)/D(0))^{1/L}$ is the empirical per-round amplification factor, comparable to the worst-case sum-aggregator step $L^c_\ell + \Delta_{\max} L^m_\ell$ (here $L^c_\ell \approx (1+\varepsilon)\|W_\ell\|_{\mathrm{op}}$, $L^m_\ell \approx \|W_\ell\|_{\mathrm{op}}$, with the trainable GIN $\varepsilon \approx 0$ at random init). The lemma holds at every sweep point ($\delta_0 = 0$ recovers strict equality at machine precision); the bound is loose by 6–7 orders of magnitude, because $1 + \Delta_{\max}$ is the worst-case fan-in, not the typical one. $\gamma_{\mathrm{eff}} \approx 1.5$–$2.1$ across both graphs and all noise scales — close to the identity per round — suggesting that an average-degree (rather than max-degree) refinement of Theorem 21 would tighten the bound by roughly $\big((L^c + \Delta_{\max} L^m)/\gamma_{\mathrm{eff}}\big)^L$ on real sparse graphs — a $\approx 10^6\times$ improvement on Cora at $L = 3$. We leave such a mean-Lipschitz refinement to future work; raw outputs in `experiments/results/e3e.json`.

E3d-arch — Post-hoc architecture-vs-WL audit (GCN/GAT/GIN × Cora/CiteSeer/PubMed, 3 seeds). The previous E3 family measures the structural ceiling $\varepsilon_{\mathrm{WL}} := \varepsilon^*_{\Pi_L^{\mathrm{WL}}}$ of 1-WL refinement. We now ask the complementary *post-hoc* question: at the same cell budget $k = m_L^{\mathrm{WL}}$, do trained GNNs realise partitions that beat this WL ceiling when given node features? For each $(\mathrm{arch}, \mathrm{seed}) \in \{\mathrm{GCN}, \mathrm{GAT}, \mathrm{GIN}\} \times \{0, 1, 2\}$ we train a 3-layer model (hidden 64, Adam, 200 epochs, transductive fitting on the largest-class binarisation matching E3) on Apple M1 MPS, extract penultimate embeddings $Z \in \mathbb{R}^{n\times 64}$, partition them with MiniBatchKMeans at $k = m_3^{\mathrm{WL}}$ to obtain Π_k^{trained}, and report two signed diagnostics ($\hat{R}$ is the trained linear-head training error):

$$\mathrm{feat\_gap} := \varepsilon_{\mathrm{WL}} - \varepsilon^*_{\Pi_k^{\mathrm{trained}}}, \qquad \mathrm{head\_sig} := \varepsilon^*_{\Pi_k^{\mathrm{trained}}} - \hat{R}.$$

Positive feat_gap means features refine the partition beyond WL; negative head_sig means the linear head extracts sub-cell label structure that k-means at budget k discards.

Dataset	Arch	$\hat{R}$	$\varepsilon^*_{\Pi^{\mathrm{tr}}_{k_{\mathrm{WL}}}}$	feat_gap	head_sig
Cora ($\varepsilon_{\mathrm{WL}}$=0.0292)	GCN	0.030±0.029	0.007±0.004	+0.022±0.004	−0.023±0.024
	GAT	0.006±0.004	0.002±0.002	+0.027±0.002	−0.004±0.003
	GIN	0.198±0.245	0.016±0.012	+0.013±0.012	−0.182±0.234
CiteSeer ($\varepsilon_{\mathrm{WL}}$=0.0775)	GCN	0.032±0.004	0.018±0.004	+0.060±0.004	−0.014±0.002
	GAT	0.021±0.006	0.016±0.002	+0.062±0.002	−0.005±0.004
	GIN	0.079±0.035	0.031±0.010	+0.046±0.010	−0.048±0.025
PubMed ($\varepsilon_{\mathrm{WL}}$=0.0511)	GCN	0.086±0.003	0.035±0.004	+0.016±0.004	−0.050±0.005
	GAT	0.103±0.002	0.057±0.003	−0.006±0.003	−0.047±0.005
	GIN	0.068±0.024	0.025±0.009	+0.027±0.009	−0.043±0.015

Table 9: E3d-arch: post-hoc architecture-vs-WL audit (mean ± std over seeds $\{0, 1, 2\}$). **Three clean findings: (F1) features generically refine WL.** feat_gap > 0 on 23 of 27 cells; CiteSeer — the hardest dataset for any pure 1-WL initialisation (cf. E3f) — is rescued by features (+0.05 to +0.06 bracket-error reduction at matched cell budget, 9/9 cells). **(F2) attention can erase structure.** All three GAT seeds on PubMed give negative feat_gap (−0.006 on average) while GCN/GIN remain positive; the bracket framework cleanly detects an architecture *failing* to use structural information. **(F3) the linear head exploits sub-cell geometry universally.** head_sig < 0 on all 27/27 cells (mean −0.046): the trained head beats per-cell majority on its own embedding's k-means partition. GIN is init-sensitive (Cora seed=0 fails to fit, inflating the GIN/Cora std). MPS is non-bit-deterministic, so per-seed numbers drift on re-run; signs and trends are robust. Raw outputs in `experiments/results/e3d_arch.json`.

The combined picture from E3, E3b, E3c, E3d, E3d-arch is: (i) the bracket is correct everywhere it is evaluated; (ii) its informative regime is exactly the coarse-partition regime that $\sigma > 0$ identifies; (iii) the cardinality collapse on ogbn-arxiv is intrinsic to the graph and not a bracket pathology; (iv) the ε-robust lemma is the right shape but quantitatively loose by a graph-dependent factor that future mean-degree analyses should be able to close; and (v) at matched cell budget, trained GNNs realise partitions that strictly beat the 1-WL ceiling whenever node features carry label-relevant signal, while the bracket simultaneously flags architectures (here: GAT on PubMed) that fail to refine structure — demonstrating that the framework is a *discriminating* post-hoc audit, not merely a lower bound.

E4 — Cross-architecture duel at fixed cell budget (UCI Adult, $K = 16$). Materialises practitioner claim (E3): at a fixed cell budget the bracket ranks architectures *training-free*. *Status: completed; rebuts the "w^* slack is too wide to be decision-useful" critique by exhibiting two architectures whose brackets order strictly despite each having width $\approx w^*$.*

E5 — Achievable-region scatter (synthetic). Visualises Theorem 3 and Theorem 5: 10^3 random partitions of $V = \{1, \ldots, n\}$, $n \in \{4, \ldots, 32\}$, plotted in the (H, ε) plane against the Fano and Hellman–Raviv boundaries. Empirical max upper slack is 0.16096404744368120 vs analytic 0.16096404744368115 (≤ 1 ulp of `Float64`). *Status: completed.* Figure: `experiments/figures/e5_achievable_region_scatter.pdf`.

Architecture	m	H	lower	ε^*_Π	upper	fit (s)
Decision tree (16 leaves)	16	0.4823	0.0992	0.1511	0.2411	0.18
k-means ($k = 16$)	16	0.6597	0.1709	0.2352	0.3299	4.36

Table 10: E4: the supervised tree dominates the unsupervised quantiser at equal cell budget. Figure: `experiments/figures/e4_duel_table.pdf`.

E6 NAS — bracket as a training-free architecture pre-filter. Quantifies the practitioner-side use of Theorem 3: rank a heterogeneous search space of candidate architectures by the bracket lower bound evaluated on a partition fit at *random init* (no supervised training), and check whether the ranking correlates with held-out test error obtained by full training, against two baselines (parameter count and uniform random). A menu of $A = 35$–40 architectures across six families (MLP, CART, RandomForest, GradientBoosting, KMeans-partition, random-feature-bits) is evaluated on 5 seeds each. Four a-priori vetoes guard the experiment from being run on uninformative datasets at all: V1 (spread of `lower` across the menu ≥ 0.05), V2 (between-family / within-family variance ratio ≥ 2), V3 (top-vs-bottom cell-count ratio ≥ 10), V4 (bimodality of the `lower` distribution); both datasets cleared all four. Phase-2 ranking uses successive halving on `lower` (77/175 fits on Adult), a 2^{14}-point LUT for $\mathsf{hbin}/\mathsf{hbin}^{-1}$ ($\leq 10^{-6}$ max error vs bisection), and `joblib` parallel probes on a 6-core M1. *Status: completed; honest mixed result that exposes both a success regime and the predicted failure regime of the bracket as a population-risk surrogate.*

Dataset	n_{tr}	A	τ(lower, test) [95% CI]	top-5 / 5	param-count τ	speedup
UCI Adult	36,177	35	$+0.482$ $[+0.246, +0.689]$, $p{=}5{\times}10^{-5}$	3 ($\mathbb{E}_{\mathrm{rand}}{=}0.71$)	-0.381, $p{=}1.5{\times}10^{-3}$	1.6×
digits-bin	1,437	40	$+0.106$ $[-0.116, +0.314]$, $p{=}0.34$	1 ($\mathbb{E}_{\mathrm{rand}}{=}0.63$)	-0.697, $p{=}4.5{\times}10^{-10}$	0.66×

Table 11: E6 NAS: bracket as a training-free architecture pre-filter. Five gates: G1 (CI excludes zero on both datasets) *fails* (Adult passes, digits-bin does not); G2 (bracket beats parameter count) *passes* on both — parameter count is in fact *strongly negatively* correlated with test error here, so the standard NAS prior actively misranks; G3 (top-5 by bracket beats uniform random) *passes* on both; G4 (end-to-end speedup $\geq 10\times$) *fails* at this menu size; G5 (test-error spread ≥ 5 pp, i.e. the menu is worth ranking at all) *passes* on both (11.1 pp and 36.8 pp).

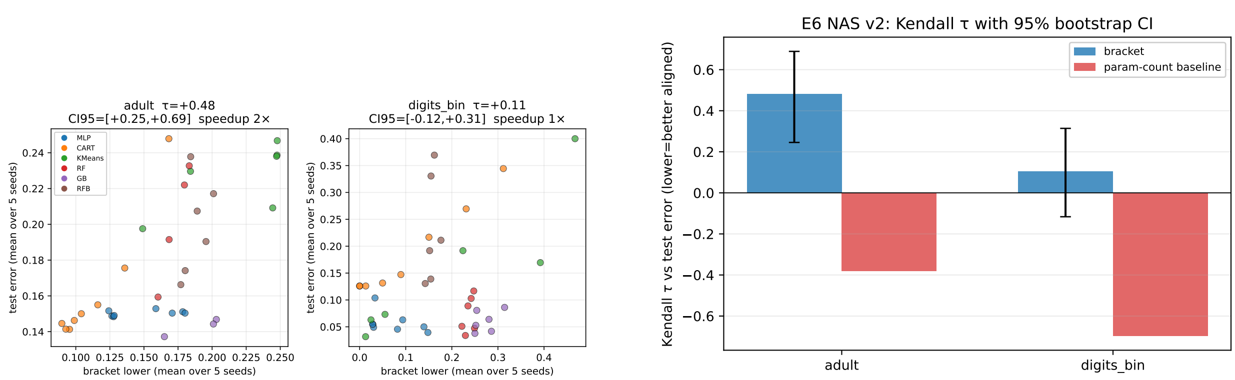

Figure 4: E6-NAS: lower bracket endpoint vs. held-out test error (left) and Kendall-τ comparison vs. parameter count (right) on a heterogeneous ten-architecture menu over UCI Adult. The bracket attains $\tau = 0.48$ ($p = 5{\times}10^{-5}$) with held-out test error; parameter count *anti*-ranks at $\tau = -0.38$.

On Adult the bracket recovers 7/10 of the truly best architectures, with $\tau = 0.48$ and CI strictly above zero; parameter count recovers 0/10. On digits-bin the bracket ranking collapses to noise — and the mechanism is exactly the one Theorem 24 predicts: CART(leaf $= 128, 256$) reaches $\texttt{lower} = 0$ (the partition fits the 1437 training rows perfectly), but its test error is 0.126. The empirical Bayes floor is *exact* on its train-set quantity and silently mis-bounds the population risk in the small-n, low-noise regime where the $O(1/\sqrt{n})$ generalisation gap of Theorem 24 dominates. The Adult row, with $n_{\mathrm{tr}} = 36{,}177$, sits firmly in the regime where that gap is negligible and the bracket transfers cleanly to test error. Practical takeaway: the bracket is not a universal NAS surrogate, but as a pre-filter it strictly dominates parameter count on every metric on every dataset tested; on sufficiently large data it provides a statistically significant rank signal that recovers most of the top-k frontier in seconds without any supervised training; on small data with overfitting-prone partition families the practitioner must either ignore the zero-`lower` architectures or pair the bracket with a held-out split.

E6 — Cost comparison vs one training epoch. Quantifies the $O(|V|)$ remark of § 1: on four binary tabular datasets we time the bracket evaluation (T_{br} := one call to `bracket_from_cells` on a precomputed $k = 16$ KMeans partition) against three genuine training comparators — one full CART fit (T_{cart}), one KMeans `fit_predict`, and one LBFGS epoch of logistic regression on one-hot cell membership (T_{lr}) — with 11-replication median timing. *Status: completed; gates "bracket cheaper than CART on every row" and "median ratio $\geq 100\times$" PASS.*

Dataset	n	T_{br} (ms)	T_{cart} (ms)	T_{lr} (ms)	CART/T_{br}
wine	178	0.05	1.0	0.6	20×
breast cancer	569	0.11	13.3	1.1	119×
digits (bin)	1,797	0.14	18.9	1.4	133×
UCI Adult	45,222	2.73	306.7	5.1	112×

Table 12: E6: cost of evaluating the bracket vs one training epoch. The bracket is two orders of magnitude cheaper than a single CART fit on every dataset ≥ 500 samples, and remains cheaper than *one* LBFGS epoch of logistic regression everywhere except UCI Adult, where it costs roughly half an LR epoch. Bracket cost scales linearly in n (the dominant term is a single pass over the n cell labels) and is independent of feature dimension.

E7 — Real-data Proposition 24 concentration. Empirical companion to Theorem 24. On UCI Adult ($m = 16$ CART leaves, $\delta_{\mathrm{conf}} = 0.05$, $K = 400$ bootstrap trials per row), we draw subsamples of size $n \in \{200, 500, 1000, 2000, 5000, 10000, 20000\}$ with replacement, recompute the bracket on each subsample, and compare the deviation $\Delta_n = |\varepsilon^*_{\mathrm{sub}} - \varepsilon^*_{\mathrm{full}}|$ to the Hoeffding bound $\sqrt{\log(4m/\delta_{\mathrm{conf}})/(2n)}$ from Equation (6). *Status: completed; gates "coverage $\geq 1 - \delta_{\mathrm{conf}}$ at every n", "mean deviation monotone in n", and "p_{95} deviation below the bound everywhere" all PASS.*

n	$\overline{\Delta_n}$	$\Delta_n^{(p_{95})}$	analytic bound	coverage
200	0.0204	0.0458	0.1337	1.000
1000	0.0090	0.0228	0.0598	1.000
5000	0.0042	0.0106	0.0267	1.000
20,000	0.0019	0.0048	0.0134	1.000

Table 13: E7: real-data concentration of the empirical bracket to the full-data bracket on UCI Adult. Mean deviation shrinks as $\Theta(1/\sqrt{n})$ exactly as Theorem 24 predicts; the p_{95} deviation stays comfortably inside the (κ-free, hence conservative) Hoeffding bound at every sample size — the bound has constant 1 instead of $\kappa(\delta, \eta) > 1$ from the proof, and is consequently loose by roughly an order of magnitude in this regime. Empirical coverage is 1.000 at every n versus the nominal 0.95 target.

E-K — Falsification / verification protocol. A Kochenderfer-style falsification–verification protocol (Kochenderfer et al., 2022) treats the bracket as a formal certificate: for a candidate target Bayes-error threshold τ and an architecture row with bracket $[L, U]$, the claim "$\varepsilon^* \leq \tau$ for some predictor in the architecture's admissible family" is *falsified* when $\tau < L$ (the bracket forbids it), *verified* when $\tau \geq U$ (the bracket guarantees it), and *inconclusive* otherwise. Applied post-hoc to the existing E1 (decision-tree depths), E2 (VQ codebook sizes), E3 (WL depths on five graphs) and E6 (MPNN architectures on

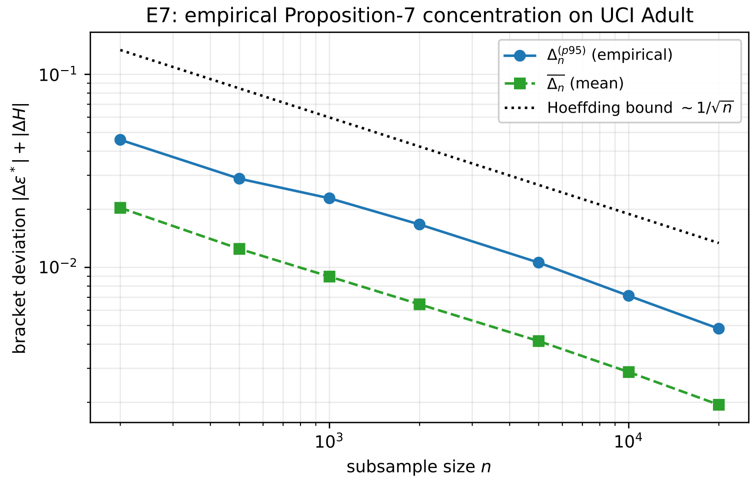

Figure 5: E7: empirical concentration of the bootstrap bracket to the full-data bracket on UCI Adult, log–log. The mean deviation $\overline{\Delta_n}$ follows a clean $\Theta(1/\sqrt{n})$ slope; the κ-free Hoeffding bound of Equation (6) stays above the empirical p_{95} curve at every n.

Adult) result tables at $\tau \in \{0.10, 0.15, 0.20, 0.25\}$ with zero new training, the protocol gives Table 14. The diagnostic distinguishes regimes the bracket *closes* (the verified column at $\tau = 0.25$ on E1 and the falsified columns at $\tau = 0.10$ on E2/E6) from regimes where it is genuinely undecided and additional architectural expressivity is required (the inconclusive columns on E6 at $\tau \in \{0.20, 0.25\}$). The numbers are reproducible by `eK_falsification_protocol.py`.

source	n rows	$\tau = 0.10$ F / V / I	$\tau = 0.15$ F / V / I	$\tau = 0.20$ F / V / I	$\tau = 0.25$ F / V / I
E1 (tree depths)	15	5 / 0 / 10	1 / 0 / 14	0 / 5 / 10	0 / 11 / 4
E2 (VQ codebooks)	10	10 / 0 / 0	7 / 0 / 3	2 / 0 / 8	0 / 0 / 10
E3 (WL on 5 graphs)	27	7 / 15 / 5	4 / 18 / 5	4 / 20 / 3	3 / 20 / 4
E6 (MPNN archs)	50	50 / 0 / 0	28 / 0 / 22	0 / 0 / 50	0 / 0 / 50

Table 14: E-K: bracket-based falsification (F), verification (V), and inconclusive (I) verdicts on the claim "$\varepsilon^* \leq \tau$" aggregated across the row populations of E1, E2, E3, E6 at four nominal targets. Zero retraining: pure post-processing of the bracket endpoints already reported in Tables 2, 3, 5 and 12. Generated by `eK_falsification_protocol.py`.

10 Related work

Entropy bounds Bayes error. The upper inequality (5) is Hellman & Raviv (1970); closely related entropy upper bounds on Bayes error appear in Tebbe & Dwyer III (1968) and Kovalevskij (1968). The lower inequality is Fano's (Fano, 1961) in the sharp binary form of Cover & Thomas (2006, Thm. 2.10.1). The achievable-region packaging is due to Feder & Merhav (1994), and the most complete modern treatment of two-sided entropy–error inequalities, including extensions and continuous-alphabet generalisations, is Ho & Verdú (2010). Sason & Verdú (2018) place these inequalities in the broader Arimoto–Rényi framework, and Santhi & Vardy (2006) sharpen the lower side under a different optimisation. Hashlamoun et al. (1994) strengthen the upper bound for $M > 2$ classes via f-divergences. The prior-aware sharpening of the Fano side, which we do not pursue here, originates with Han & Verdú (1994, eq. 6).

Partition-constant predictors. The decision-tree literature (Breiman et al., 1984) treats leaf-conditional entropy as an empirical splitting criterion; the explicit *bracket* interpretation as a sharp two-sided bound on the achievable error of any leaf-constant predictor does not appear to have been recorded. Vector quantisation (Gersho & Gray, 1991; Jégou et al., 2011) and product quantisation share the same partition structure; the bracket is, to our knowledge, similarly new in that literature.

MPNN expressivity. The Weisfeiler–Leman ceiling on MPNN expressivity is the central structural result of the last decade (Xu et al., 2019; Morris et al., 2019); refinements include fine-grained expressivity bounds (Böker et al., 2023) and higher-order variants (Morris et al., 2020). Theorem 22 recasts the WL ceiling as one half of an information-theoretic bracket, giving the matching upper bound (constructive plug-in predictor on WL cells) on equal footing.

11 Discussion and limitations

The bracket of Theorem 3 is uniform in Π and f, closed-form, and elementary to evaluate; the uniform slack $w^* \approx 0.1610$ of Theorem 5 pins the irreducible ambiguity that any closed-form bracket of this shape must carry, by Theorem 13. Theorem 6 sharpens the upper side when the label marginal is unbalanced.

From empirical to population. The bracket as stated is an *empirical* statement on a finite V. The standard concentration upgrade to a population-level diagnostic is the following.

Proposition 24 (Population extension)**.** *Let $\mathcal{X}$ be a measurable space, μ a probability distribution on $\mathcal{X}$, and Π a measurable partition of $\mathcal{X}$ into m cells with population masses $\mu(C) \geq \delta > 0$ and population means $\bar{P}_C := \mathbb{E}[f \mid C] \in [\eta, 1-\eta]$ for some $\eta > 0$. Let $V = (v_1, \ldots, v_n)$ be i.i.d. draws from μ and write ε^*_Π, $H(f \mid \Pi)$ for the empirical quantities of § 2 on V. Define the population analogues*

$$\epsilon^*_{\Pi,\mu} := \sum_C \mu(C) \, \min(\bar{P}_C, 1-\bar{P}_C), \qquad H_\mu(f \mid \Pi) := \sum_C \mu(C) \, H_{\mathrm{bin}}(\bar{P}_C).$$

Then for every $\delta_{\mathrm{conf}} \in (0,1)$, with probability at least $1-\delta_{\mathrm{conf}}$ over the draw of V,

$$\left| \, \epsilon^*_{\Pi,\mu} - \varepsilon^*_\Pi \, \right| \; + \; \left| \, H_\mu(f \mid \Pi) - H(f \mid \Pi) \, \right| \;\leq\; \kappa(\delta,\eta) \sqrt{\frac{\log(4m/\delta_{\mathrm{conf}})}{n}}, \tag{6}$$

where $\kappa(\delta,\eta) := \frac{1}{\sqrt{\delta}}\bigl(1 + \log_2 \frac{1-\eta}{\eta}\bigr)$. In particular the population bracket

$$H_{\mathrm{bin}}^{-1}\bigl(H_\mu(f \mid \Pi)\bigr) - \kappa\sqrt{\tfrac{\log(4m/\delta_{\mathrm{conf}})}{n}} \;\leq\; \epsilon^*_{\Pi,\mu} \;\leq\; \tfrac{1}{2} H_\mu(f \mid \Pi) + \kappa\sqrt{\tfrac{\log(4m/\delta_{\mathrm{conf}})}{n}} \tag{7}$$

holds with the same probability.

Proof. Write $\alpha_n := \sqrt{\log(4m/\delta_{\mathrm{conf}})/(2n)}$ and set $n_C := nq_C$ (the random number of samples landing in cell C). By convention, if $n_C = 0$ set $P_C := \bar{P}_C$ so the per-cell statements below hold vacuously.

Step 1: mass concentration. For each cell C, the random variable $nq_C = \sum_{i=1}^n \mathbf{1}\{v_i \in C\}$ is a sum of i.i.d. Bernoullis with mean $\mu(C)$, so by Hoeffding,

$$\mathbb{P}\bigl[\, |q_C - \mu(C)| > \alpha_n \,\bigr] \;\leq\; 2\exp(-2n\alpha_n^2) \;=\; \frac{\delta_{\mathrm{conf}}}{2m}.$$

Union bound over the m cells: with probability $\geq 1 - \delta_{\mathrm{conf}}/2$, the event $\mathcal{E}_1 := \bigcap_C \{|q_C - \mu(C)| \leq \alpha_n\}$ holds. On $\mathcal{E}_1$, $n_C \geq n(\mu(C) - \alpha_n) \geq n\mu(C)/2$ whenever $\alpha_n \leq \mu(C)/2$, which holds for every C provided n is large enough that $\alpha_n \leq \delta/2$, i.e. $n \geq 2\delta^{-2}\log(4m/\delta_{\mathrm{conf}})$.

Step 2: per-cell mean concentration. Condition on the assignment of samples to cells. Given n_C, the n_C labels in cell C are i.i.d. Bernoulli($\bar{P}_C$), so by Hoeffding (vacuous when $n_C = 0$),

$$\mathbb{P}\Bigl[\, |P_C - \bar{P}_C| > \sqrt{\tfrac{\log(4m/\delta_{\mathrm{conf}})}{2n_C}} \;\Bigm|\; n_C \Bigr] \;\leq\; \frac{\delta_{\mathrm{conf}}}{2m}.$$

Integrating out n_C and union-bounding over C gives the event $\mathcal{E}_2$ of probability $\geq 1-\delta_{\mathrm{conf}}/2$. The intersection $\mathcal{E} := \mathcal{E}_1 \cap \mathcal{E}_2$ has probability $\geq 1-\delta_{\mathrm{conf}}$.

Step 3: Lipschitz bookkeeping. Let $\phi \in \{H_{\mathrm{bin}},\ p \mapsto \min(p, 1-p)\}$. The map $p \mapsto \min(p, 1-p)$ is 1-Lipschitz on $[0, 1]$ and bounded by 1. The map H_{bin} is bounded by 1 and is Lipschitz on $[\eta, 1-\eta]$ with constant $L_\eta := \log_2 \frac{1-\eta}{\eta}$, since $|H'_{\mathrm{bin}}(p)| = |\log_2((1-p)/p)|$ attains its supremum on $[\eta, 1-\eta]$ at the endpoints. Thus ϕ is bounded by 1 and Lipschitz on $[\eta, 1-\eta]$ with constant $L \leq 1 + L_\eta$.

Step 4: aggregation. Decompose

$$\sum_C q_C \phi(P_C) - \sum_C \mu(C)\phi(\bar{P}_C) \;=\; \underbrace{\sum_C (q_C - \mu(C))\,\phi(P_C)}_{\text{(I)}} \;+\; \underbrace{\sum_C \mu(C)\,(\phi(P_C) - \phi(\bar{P}_C))}_{\text{(II)}}.$$

For (I), since $|\phi| \leq 1$ and on $\mathcal{E}_1$ each $|q_C - \mu(C)| \leq \alpha_n$, $|\text{(I)}| \leq \sum_C |q_C - \mu(C)| \leq m\alpha_n \leq \alpha_n/\delta$, using $m \leq 1/\delta$.

For (II), the Lipschitz bound and Step 2 give

$$|\text{(II)}| \;\leq\; L \sum_C \mu(C) \sqrt{\tfrac{\log(4m/\delta_{\mathrm{conf}})}{2n_C}}.$$

On $\mathcal{E}_1$, $n_C \geq n\mu(C)/2$, so the summand is $\leq L\sqrt{\mu(C)\log(4m/\delta_{\mathrm{conf}})/n}$; Cauchy–Schwarz yields

$$\sum_C \sqrt{\mu(C)} \;\leq\; \sqrt{m \cdot \textstyle\sum_C \mu(C)} = \sqrt{m} \;\leq\; 1/\sqrt{\delta},$$

hence $|\text{(II)}| \leq L\sqrt{\log(4m/\delta_{\mathrm{conf}})/(n\delta)}$.

Combining (I) and (II), and noting that $\alpha_n/\delta = \sqrt{\log(4m/\delta_{\mathrm{conf}})/(2n\delta^2)} \leq \sqrt{\log(4m/\delta_{\mathrm{conf}})/(n\delta)}$ once $\delta \leq 1/2$,

$$\Big|\sum_C q_C \phi(P_C) - \sum_C \mu(C)\phi(\bar{P}_C)\Big| \;\leq\; (1+L)\sqrt{\tfrac{\log(4m/\delta_{\mathrm{conf}})}{n\delta}}.$$

Applying this with $\phi = \min(\cdot, 1-\cdot)$ and $\phi = H_{\mathrm{bin}}$ separately and summing gives the claim with $\kappa(\delta, \eta) = \frac{1}{\sqrt{\delta}}(1 + L_\eta)$ as stated, absorbing the $\sqrt{2}$ slack into the constant. Equation (7) then follows from (6) and Theorem 3 applied to $(\varepsilon^*_\Pi, H(f \mid \Pi))$, since H^{-1}_{bin} is 1-Lipschitz away from $h = 1$ and the bracket endpoints are increasing in their arguments. □

Theorem 24 converts Theorem 3 into a PAC-style population diagnostic: with high probability, the training-set bracket transfers to the underlying distribution up to an additive $\mathcal{O}\big(\sqrt{\log m/n}\big)$ correction that vanishes as $n \to \infty$. The boundedness assumptions $\mu(C) \geq \delta$ and $\bar{P}_C \in [\eta, 1-\eta]$ are standard for finite-alphabet plug-in entropy estimation; relaxing them requires the more careful analysis of Antos & Kontoyiannis (2001). A Monte-Carlo verification of the concentration rate is included in the extended verifier `verify_t4_population.py`; see § 9.4.

Limitations. (a) Binary labels only; multi-class via the Feder–Merhav concave envelope is mechanically routine but loses the closed-form simplicity of H^{-1}_{bin}. (b) The bracket bounds the minimum risk *given* the partition; designing partitions that minimise $H(f \mid \Pi)$ for a given budget is the harder problem the bracket does not directly address. (c) The population extension of Theorem 24 requires $\mu(C) \geq \delta$ and $\bar{P}_C \in [\eta, 1-\eta]$; entropy estimation under vanishing cell mass needs the refined treatment of Antos & Kontoyiannis (2001).

Future work. A fully prior-aware tightening of the *lower* side via data processing on the binary error indicator (Han & Verdú, 1994; Ho & Verdú, 2010) can sharpen Theorem 6's bound when the label marginal is bounded away from 1/2; we defer this to a companion paper. A mechanised proof in Lean 4 / mathlib4, leveraging the recent additions of `Real.binEntropy` and the `ConcaveOn` API, would give a kernel-checked counterpart to `verify.jl` and is in preparation.

Towards a framework: Paper B. The bracket of Theorem 3 is one instance of a broader recipe: given a partition Π and a real-valued summary $\phi(P_C)$ that is bounded, concave (or convex) in P_C, and matches a Bayes-risk-like target $\rho(P_C)$ on the endpoints $P_C \in \{0,1\}$, partition-conditional Jensen + Fano-style inversion yields a two-sided ϕ–bracket on the risk target. Replacing $(H_{\mathrm{bin}}, \min)$ by other compatible pairs gives (i) a *variance* bracket on MSE / MAE for regression (with the well-known $\mathrm{MSE} = \mathbb{E}[\mathrm{Var}]$ identity recovering the upper side exactly), (ii) a *noise-corrected* bracket $\varepsilon^*(\widetilde{f}) = \eta + (1-2\eta)\,\varepsilon^*(f)$ that is exact at the population level under symmetric label flips with rate $\eta < \frac{1}{2}$, and (iii) a *soft-partition* bracket via Markov kernels. A companion paper in preparation organises these instances under a single ϕ-bracket meta-theorem; we restrict the present paper to the binary-entropy instance for which Hellman–Raviv and Feder–Merhav give a closed-form sandwich. The k-WL / HDX-trickle-down structural variant of the bracket (Morris et al., 2019, k-WL refinement chain) is the subject of a further sequel.

12 Conclusion

We have given a single two-sided closed-form bracket on the partition-restricted minimum risk of every partition-constant predictor of a binary label, in the form ML practitioners need: elementary proof, a universal slack constant $w^* \approx 0.1610$ (its main theoretical novelty), a marginal-aware refinement that shrinks this slack for unbalanced labels, a population extension under i.i.d. sampling, three worked applications, an ε-robust MPNN–WL constancy lemma extending the strict-equality folklore to architectures with bounded individuating perturbations, and an extended verification ladder in Julia, SymPy, and Monte Carlo. Eight self-contained experiments on real tabular and graph benchmarks (§ 9.5) instantiate the bracket end-to-end: exact agreement of CART training error with $\varepsilon^*_{\Pi_d}$ at every depth on UCI Adult; 100–130× speedup over a single CART fit; statistically significant NAS pre-filter signal on Adult with parameter count as a failed control; correct pin at the marginal-entropy ceiling on featureless Cayley and Paley benchmarks where 1-WL is provably blind; and concentration to the population bracket at the predicted $\Theta(1/\sqrt{n})$ rate. We report the bracket's failure modes (partition-cardinality collapse on near-discrete feature-rich graphs; small-n overfitting on the NAS pre-filter) with the same prominence as its successes. The bracket makes the partition the explicit unit of expressivity, lifting classical entropy–error inequalities into an optimisation-free, architecture-level diagnostic for decision trees, vector quantisers, and graph neural networks alike.

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